A Framework for Auditing Value Sensitivity in Large Language Model–Assisted Health Policy Simulation

Yichao Jin *
EPPS, University of Texas at Dallas

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Abstract

Objective: To test whether LLM-assisted policy simulations are robust to model choice, prompt framing, and measured moral foundations, using administrative burden in U.S. vaccination access as a testbed.
Study Setting and Design: Multi-level diagnostic framework. Fifteen LLM endpoints completed an MFQ-30–derived probe and simulated vaccination willingness for 27 standardized profiles under low- and high-burden conditions, yielding tests of composite orientation (H1), rationale-frame association (H2), and within-model frame manipulation (H3).
Data Sources and Analytic Sample: Responses were collected on 16–17 September 2026 (UTC) from 13 LLM endpoints accessed through provider APIs and gateways. H1: 15 model endpoints; H2: 2,019 analyzable low-high repetition pairs nested within 405 model-profile cells; H3: 108 clusters (2,159 observations). The 27 synthetic profiles are constructed test cases; no human participants were involved.
Principal Findings: (1) A composite derived from an MFQ-30–derived probe does not significantly predict cross-model variation in simulated burden effects ($\beta_1 = -6.69$, 95% CI $[-16.27, +2.88]$, $p = 0.155$, $R^2 = 0.149$): higher individualizing-composite models show no consistent pattern in burden effects. (2) Among the two rationale frames with sufficient variation for estimation, neither showed a detectable within-pair association with simulated burden effects (BH-adjusted $p \geq 0.86$). (3) Assigned frames shift simulated burden effects within models: autonomy +4.96 [3.57, 6.35], collective obligation −10.88 [−12.54, −9.21], equity/access +4.84 [3.45, 6.24]; the autonomy direction reverses in one of four models.
Conclusions: In this testbed, LLM-assisted policy simulation is frame-sensitive: identical models, profiles, and scenarios yield materially different estimates (pooled shifts of 4.8–10.9 points on a 0–100 scale) depending on analyst-selected framing. These results characterize simulation behavior, not real-world vaccination response.

What is known on this topic

- LLMs are increasingly used to simulate social judgments and policy effects in health services research.

*University of Texas at Dallas,Yichao.Jin@UTDallas.edu

- Model values are usually measured with survey prompts rather than downstream simulated policy effects, so whether those orientations shape simulated effects is unknown.

What this study found

- A composite derived from an MFQ-30–derived probe does not significantly predict cross-model variation in simulated burden effects: higher individualizing-composite models show no consistent pattern in burden effects.
- Naturally expressed rationale frames show no detectable within-pair association with simulated effects under the primary specification.
- Assigned frames shift simulated effects within models: collective obligation reduces them in all four tested models; autonomy and equity increase them in three.

Keywords: large language models, simulation robustness, administrative burden, policy simulation, frame sensitivity, calibration

1 Introduction

Administrative burden in U.S. health services—the learning, psychological, and compliance costs of interacting with public programs—is a well-documented barrier to care [1, 2]. It arises from forms, documentation, waiting time, procedural uncertainty, and perceived loss of autonomy, and it falls hardest on people whose informational and material resources are already constrained [1, 3–5]. Policymakers therefore need tools that anticipate the access effects of procedural changes before those changes are implemented [6, 7].

Large language models (LLMs) are increasingly used as synthetic participants and behavioral simulators—as substitutes for survey respondents, as simulated agents, and as predictors of experimental outcomes [8–14]—including in health applications such as synthetic quality-of-life survey responses and simulation of population responses to policy interventions [15–17]. As these methods move from measuring opinions to predicting policy-relevant effects, a new methodological problem arises: seemingly technical simulated policy-effect estimates may depend on analyst-selected normative framing. If normative-frame sensitivity propagates to policy simulation, otherwise identical model–profile scenarios could yield different simulated effects under alternative governance-value frames.

That LLM outputs respond to prompt wording and framing is well established: survey-based value and opinion measurements shift with response format, prompting technique, and phrasing [18–21]; one evaluation varies 62 value propositions across 420 prompt variations (156,000 responses) and finds systematic shifts in measured positions [22]. What remains unclear is whether this sensitivity propagates to the quantities policy analysts consume—simulated marginal effects of a policy condition. The distinction matters: for questionnaire-based value measurement, prompt sensitivity degrades an instrument; for an HSR simulation, a shift of several points in a simulated burden effect changes the policy reading of the estimate. Our contribution is to relocate this finding from stated opinions to simulated policy effects and to make it measurable through a framework that separates aggregate orientation, expressed rationale frames, and experimentally assigned frame clauses.

This paper develops and evaluates a multi-level diagnostic framework for auditing whether LLM-assisted policy simulation outputs are robust to alternative normative framings. We use administrative burden in U.S. vaccination access as a controlled HSR testbed to demonstrate the framework.

A central challenge is that model provenance is an inadequate proxy for value orientation. To avoid treating development origin as a fixed ideological category, we independently measure each model's moral foundations using an MFQ-30–derived probe adapted from the 30-item Moral Foundations Questionnaire, capturing five foundations: care, fairness, loyalty, authority, and sanctity [23]. We use MFQ-30 rather than the newer MFQ-2 to maintain comparability with prior LLM value-probing literature and the established five-foundation theoretical framework.

The study addresses three research questions. First, do LLMs with different measured moral-foundations profiles (MFQ-30) produce different simulated marginal effects of administrative burden on vaccination willingness? Second, are simulated burden effects systematically associated with the interpretive frames used in model rationales? Third, can simulated burden effects be shifted within the same base model by experimentally assigning governance-value frames?

Contributions: (1) shift LLM value-sensitivity study from opinion to simulated policy effects; (2) apply an MFQ-30–derived probe using the original two-part 0–5 administration to measure LLM values; (3) provide a multi-level diagnostic framework for LLM simulation robustness.

2 Theoretical Framework and Hypotheses

2.1 Moral Foundations

Moral foundations theory proposes that moral judgments are built on innate, culturally elaborated foundations: care/harm, fairness/cheating, loyalty/betrayal, authority/subversion, and sanctity/degradation [23, 24]. The theory has been used to characterize ideological differences in human populations: individualizing foundations (care, fairness) emphasize protection of individuals, whereas binding foundations (loyalty, authority, sanctity) emphasize the integrity of groups and institutions. We measure each model's moral profile with the 30-item Moral Foundations Questionnaire (MFQ-30), an MFQ-30–derived probe adapted from the standard instrument in this literature. Prior work has used the same instrument to probe LLM moral judgments and found that model responses track the human-like structure of moral foundations [25, 26]. Reported value and bias profiles also depend on how the instrument is administered: constrained versus open-ended response formats, prompting technique, and survey wording yield divergent measurements [18–20], and political-orientation estimates vary across model families and instruments [27–30].

2.2 Three-Level Diagnostic Framework

The framework proceeds from coarse to fine-grained diagnostics. Each diagnostic is informative regardless of the others.

Diagnostic 1 — Aggregate Orientation. We first test whether a model-level moral-foundations composite (individualizing minus binding foundations) predicts cross-model variation in simulated burden effects. Because the composite is derived from an MFQ-30–derived probe based on a human-validated questionnaire, a null result here would be informative: it would indicate that even an MFQ-30–derived scalar moral score carries little cross-model audit signal for simulated policy effects, suggesting that scalar aggregate orientation scores provide limited evidence as a standalone cross-model audit signal in this endpoint sample.

Diagnostic 2 — Naturally Expressed Framing. We examine whether interpretive frames expressed in model rationales systematically co-occur with simulated burden effects. Access-barrier, autonomy-infringement, and coercive-backlash frames should correspond to larger effects; procedural-legitimacy and collective-responsibility frames to smaller effects. Because frames are generated rather than assigned, this diagnostic identifies rationale–outcome co-expression rather than causal pathways, and we test whether any such co-expression survives model-by-profile pair fixed effects.

Diagnostic 3 — Experimental Frame Sensitivity. We test whether an experimentally assigned governance-value prompt clause causally shifts simulated burden effects within the same base model. The manipulated object is the prompt clause itself: the causal claim therefore concerns the assigned framing of the prompt, not any claim about the values a model holds internally. Autonomy and equity/access frames should increase

effects; collective-obligation frames should decrease them. Holding model, profiles, and scenarios fixed attributes any shift to framing itself.

Each diagnostic addresses a distinct aspect of prompt-frame sensitivity and contributes to the overall assessment of simulation robustness.

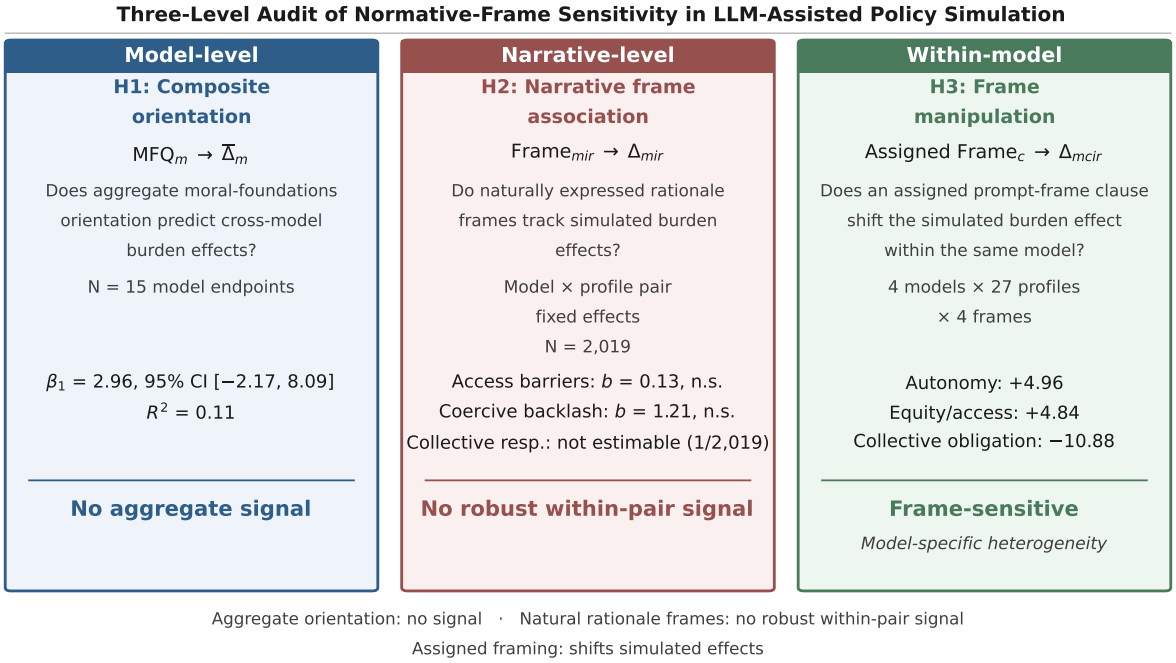


Figure 1: Three-level audit of normative-frame sensitivity. H1 tests whether a model-level MFQ composite predicts cross-model variation in administrative-burden effects (N = 15 endpoints). H2 tests whether narrative frames expressed in model rationales are associated with simulated burden effects under model-by-profile pair fixed effects (N = 2,019). H3 tests whether an experimentally assigned governance-value prompt clause shifts the simulated burden effect within the same base model (4 models × 27 profiles × 4 frames). The three diagnostics are independent components: each is informative regardless of the outcome of the others.

3 Experimental Design and Methods

3.1 Model Sample

We evaluate thirteen LLM endpoints spanning nine provider ecosystems (OpenAI, Anthropic, Google, Meta, Mistral, Cohere, Moonshot AI, DeepSeek, Qwen). The sample includes frontier proprietary, lower-cost, hosted open-weight, enterprise-oriented, and non-U.S. models. All models were run with temperature = 0 where supported; each cell repeated five times. All responses analyzed here were collected between 16 and 17 September 2026 (UTC). Detailed identifiers are reported in the Online Supplement.

3.2 Moral-Foundations Battery

We administered the 30-item Moral Foundations Questionnaire (MFQ-30), an MFQ-30–derived probe based on the standard instrument in moral psychology [23], measuring

five foundations: care, fairness, loyalty, authority, and sanctity. Each foundation uses six items (three relevance-judgment items on a 0–5 scale and three agreement items on a 0–5 scale), following the original two-part administration. Following prior LLM-probing work with this instrument [25, 26], we compute the moral-foundations composite as the unweighted signed sum of standardized foundation scores: $\mathrm{MFQ}_m = Z_m^{(\mathrm{CA})} + Z_m^{(\mathrm{FA})} - Z_m^{(\mathrm{LO})} - Z_m^{(\mathrm{AU})} - Z_m^{(\mathrm{SA})}$, where care and fairness (individualizing foundations) enter positively and loyalty, authority, and sanctity (binding foundations) enter negatively. All thirteen endpoints were administered the identical frozen battery protocol in the canonical rerun; the system prompt, item template, output schema, and their SHA256 hashes are reported in Appendix C.

3.3 Administrative-Burden Simulation

To measure each model's simulated burden effect, we designed a vaccination-access simulation as a controlled HSR testbed, in which the same standardized profile faces two policy conditions: low administrative burden and high administrative burden. The simulation was administered after the orientation battery but before any frame manipulation, using a fresh API call for each model–profile–condition–repetition.

3.3.1 Standardized synthetic profiles

We constructed 27 standardized synthetic profiles as controlled test cases that cross income, work flexibility, and health-care access while holding age, gender, residence, education, family status, and trust in public health authorities constant. They are experimental stimuli designed to isolate how models interpret the same scenarios, not a representation of any human population; the full set is in the Online Supplement.

3.3.2 Low- and high-burden scenarios

The two conditions differ only in administrative requirements: low burden (online scheduling, nearby pharmacy, minimal documentation, about 15 minutes) and high burden (in-person registration, income verification, proof of residency, paper forms, a 1–2 hour wait, one-week advance appointment). These are a **stylized testbed** for auditing simulation robustness, not estimates of real-world burden magnitude.

3.3.3 Outcome measure

For each model, profile, burden condition, and repetition we obtained a predicted willingness on a 0–100 scale from a structured JSON response (integer score plus a brief rationale). The primary outcome is the within-profile burden effect $\Delta_{mir} = W_{mi,\mathrm{low},r} - W_{mi,\mathrm{high},r}$, positive when higher burden lowers predicted willingness; model-level analyses average across profiles and repetitions.

3.4 Task-Compliance Diagnostics

Because comparable benchmark scores are not uniformly available for all fifteen endpoints, we do not construct a composite capability score. Instead we report task-specific diagnostics computed from the study's own outputs—JSON validity, score-range compliance,

non-refusal, repetition-level stability, burden-effect dispersion, and flip rate (Table S1 in the Online Supplement)—which verify protocol compliance rather than capability.

3.5 Narrative coding framework

To test H2, rationales were coded with five theoretically derived frames (autonomy infringement, procedural legitimacy, collective responsibility, access barriers, coercive backlash) using a rule-based algorithm with predefined regular-expression patterns; frames are not mutually exclusive. A manual coding audit of 200 rationales (about 10% of the high-burden sample), in which the author reviewed rationale text alongside the algorithm's frame assignments and checked every positive assignment against the frame definitions, identified no systematic coding errors. Because that audit was not independent of the algorithm's labels and could not bound false negatives, the frame indicators are used as descriptive evidence of co-expression rather than as validated measurements. Patterns are in the Online Supplement.

3.6 Within-Model Frame Manipulation

To test H3, we varied system-prompt frames within the same base model while holding profiles, scenarios, and output format fixed. Only the frame clause changes, so the experiment identifies the causal effect of assignment to the specified prompt-frame condition on the model-generated burden effect within a model; it does not speak to the values a model is claimed to hold.

3.6.1 Model selection

The experiment used a purposive four-model subset (GPT-4.1, Llama 3.1 70B Instruct, Qwen3.7 Plus, Mistral Large) covering a frontier proprietary model, a hosted open-weight model, a non-U.S. ecosystem, and a low-composite case. This tests whether frame sensitivity can be induced across diverse endpoints, not its distribution over all fifteen.

3.6.2 Frame conditions and randomization

Four system-prompt frames were used: *neutral*, *autonomy*, *collective obligation*, and *equity/access* (weighting individual choice, civic responsibility, and unequal procedural costs, respectively). The manipulation was fully crossed—every model evaluated every profile under every frame—with frame order randomized across runs.

3.6.3 Statistical analysis

The frame effect is estimated with model and profile fixed effects and frame indicators (neutral as reference), with Holm–Bonferroni adjustment. Because the manipulation is fully crossed within models, profile fixed effects absorb time-invariant profile heterogeneity, and standard errors are clustered on the 108 model–profile units. The inferential target is the within-model–profile assigned-frame contrast; because only the prompt clause is manipulated, the estimate is the causal effect of assignment to the specified prompt-frame condition on the model-generated burden effect rather than an effect of model-held values.

3.7 Estimation Strategy

In H1 the MFQ composite varies only at the model level, so that analysis is descriptive; In H2, the primary specification uses model-by-profile pair fixed effects so that frame coefficients are identified from repetition-level variation within the same model–profile cell.

3.7.1 Orientation-effect models (H1)

The primary H1 analysis tests whether models with higher moral-foundations composite scores predict larger burden effects:

$$\bar{\Delta}_m = \beta_0 + \beta_1 \text{MFQ}_m + \epsilon_m,$$

where the MFQ composite is the signed sum of standardized foundation scores: each foundation score is the unweighted mean of its six items, each standardised across endpoints, and the composite is $\text{MFQ}_m = Z(\text{care})_m + Z(\text{fairness})_m - Z(\text{loyalty})_m - Z(\text{authority})_m - Z(\text{sanctity})_m$ (Appendix C); because its variation is model-level, this is a descriptive cross-model association rather than a high-powered population estimate. We also fit each of the five foundations separately as a diagnostic check.

3.7.2 Narrative frame-decomposition models (H2)

H2 estimates a frame–outcome association model with one fixed effect per model–profile pair:

$$\Delta_{mir} = \alpha_{mi} + \sum_k \beta_k \mathbb{I}(k \in r_{mir}) + \varepsilon_{mir},$$

where α_{mi} is an indicator for the model–profile pair (405 pairs in the fifteen-model sample). The frame indicators are rationale-level and vary across the five repetitions of a pair, so the coefficients are identified exclusively from repetition-level changes in frame presence within the same model–profile pair. Pair fixed effects absorb the cell-specific baseline, including any model-specific response to particular profiles that separate model and profile fixed effects would leave in the error term; a specification with separate model and profile fixed effects (additive heterogeneity only) is reported as a sensitivity analysis in Appendix H. The MFQ composite does not enter the H2 model. Estimates are co-expression, not causal mediation, because willingness and rationales are generated in the same response; standard errors are clustered at the model–profile level and frame-specific p-values use Benjamini–Hochberg FDR at $q = 0.10$.

3.7.3 Frame-manipulation models (H3)

H3 uses a two-way fixed-effects model: $\Delta_{mcir} = \alpha_m + \mu_i + \sum \beta_k \mathbb{I}(c = k) + \varepsilon_{mcir}$, with neutral as reference. Pairwise comparisons use Holm–Bonferroni correction; standard errors are clustered at the model–profile level.

3.7.4 Multiple-testing and robustness checks

Multiple-testing adjustments: Benjamini–Hochberg FDR ($q = 0.10$) for H2, Holm–Bonferroni for H3, none for H1 (single coefficient).

3.8 Human Subjects and Institutional Review

This study did not involve human subjects: all analytic inputs are synthetic profiles constructed by the author and text outputs generated by commercial LLM APIs. No human participants were recruited or observed, and no identifiable or protected health information was accessed. The work therefore does not constitute human-subjects research, institutional review board approval was not required, and no IRB approval or exemption number applies. Model outputs were obtained under the providers' standard terms of service, and no model was fine-tuned or trained on study inputs.

3.9 Data and Code Availability

Replication materials—the simulation and frame-manipulation runners, the orientation battery and its scoring code, the frame-coding patterns, the analysis scripts, and the derived response files—are available at `https://github.com/yichao2022/value-sensitivity-llm-audit` under the MIT License (Copyright 2026 Yichao Jin), which permits use, modification, and redistribution with attribution. Raw model responses remain subject to the providers' terms of service. All reported results can be reproduced from the archived response files and committed scripts.

4 Results

4.1 Data Quality and Descriptive Statistics

Table S1 in the Online Supplement reports task-compliance diagnostics for the expanded fifteen-model simulation (4,050 raw responses; 2,025 possible model–profile–repetition pairs, of which 2,019 are analyzable because six high-burden responses from Gemini 2.5 Flash failed structure checks). JSON validity, score-range compliance, and non-refusal rates were 100% for fourteen of fifteen models; Gemini 2.5 Flash had six malformed responses (97.8% compliance). The flip rate was 0% for every model, indicating that the high-burden condition moved mean predicted willingness in the expected direction in every profile; this metric documents directional consistency of the simulated responses and does not by itself establish that models understood the manipulation. An auxiliary burden-classification diagnostic fielded for that purpose returned the correct burden label in all 810 scenarios (100%, exact 95% CI 99.5–100; Appendix F), so recognition of the assigned condition is directly supported; because that check was administered separately from the simulation runs and uses the same scenario texts, it speaks to comprehension rather than to the magnitude of the simulated effects. Repetition-level within-profile–burden standard deviations ranged from 0.03 to 3.14 points on the 0–100 scale, indicating high within-model stability. Mean predicted willingness varied substantially across models (47.99–65.48), motivating the use of the within-profile marginal burden-effect outcome in the main analyses.

4.2 Orientation Profiles and Predicted Burden Effects

Table 1 reports each model's measured moral-foundations profile together with predicted willingness under low- and high-burden conditions. Mean burden effects ($\bar{\Delta}$) range from

23.07 points (GPT-4o) to 54.45 points (Qwen3.7 Plus) across models. The MFQ composite summarizes whether a model's profile is more individualizing (positive) or more binding (negative); predicted willingness is lower under high-burden conditions for every model. Baseline willingness also varies substantially across models, which is why the analyses use the within-profile burden effect rather than raw willingness levels.

Table 1: Moral-foundations profiles and predicted administrative-burden effects by model

Model	MFQ	Low W	High W	$\bar{\Delta}$	95% CI
Qwen3.7 Plus	-3.44	83.33	28.87	54.45	[51.45, 57.45]
Gemini 2.5 Pro	3.93	77.44	23.52	53.93	[50.93, 56.93]
Gemini 2.5 Flash	2.32	81.63	33.48	48.15	[45.15, 51.15]
Kimi K2.6	1.59	74.31	36.19	38.13	[35.13, 41.13]
DeepSeek-V3.2	1.83	75.76	37.96	37.79	[34.79, 40.79]
Mistral Large	-0.73	76.00	39.04	36.96	[33.96, 39.96]
Claude Opus 4.8	0.32	70.44	34.63	35.81	[32.81, 38.81]
Cohere Command R+	-3.94	76.78	41.04	35.74	[32.74, 38.74]
Claude Sonnet 4.6	-0.10	63.73	32.24	31.50	[28.50, 34.50]
GPT-4.1 mini	-1.55	73.48	44.15	29.33	[26.33, 32.33]
Llama 3.1 70B Instruct	1.34	80.00	50.96	29.04	[26.04, 32.04]
GPT-4.1	4.75	74.67	49.04	25.63	[22.63, 28.63]
GPT-4o	0.25	76.33	53.26	23.07	[20.07, 26.07]

Notes: MFQ = signed sum of standardized foundation scores: care + fairness − loyalty − authority − sanctity. Low W and High W are mean predicted vaccination willingness scores under low- and high-burden conditions. $\bar{\Delta}$ = Low W − High W. 95% CI calculated as $\bar{\Delta} \pm 1.96 \times SE$, where SE is the standard error of the 27 profile-level burden effects (each profile-level effect is the mean over that profile's repetitions), so repetitions are not treated as independent observations.

4.3 Hypothesis Tests

H1 tests whether the model-level MFQ composite predicts cross-model variation in average burden effects; H2, whether rationale-level frames co-vary with predicted effects; and H3, whether assignment to a specified prompt-frame condition causally shifts predicted effects within a model. The three tests are independent: each addresses a distinct level of evidence and is informative regardless of the others.

4.3.1 H1: Orientation–Effect Models

H1 Result: Aggregate orientation does not significantly predict cross-model variation. The MFQ composite does not significantly predict cross-model variation in simulated burden effects ($\hat{\beta} = -0.19$, 95% CI [−2.50, +2.12], $p = 0.875$, $R^2 = 0.002$). Models with higher individualizing-composite orientations show no consistent pattern in simulated burden effects. Per-foundation diagnostics show mixed point estimates across binding foundations.

Figure 2 shows the MFQ composite against mean burden effects: the fitted line is shallow, and models with more individualizing (higher-composite) orientations do not

systematically produce smaller effects. Per-foundation diagnostics are consistent with the aggregate null: none of the five foundations reached conventional significance. All three binding-foundation point estimates were negative, with loyalty showing the largest magnitude ($\hat{\beta} = -4.73$, $p = 0.095$), followed by authority ($\hat{\beta} = -4.25$, $p = 0.138$) and sanctity ($\hat{\beta} = -3.35$, $p = 0.250$).

The null result for a composite derived from an MFQ-30–derived probe based on a human-validated questionnaire indicates no statistically detectable cross-model association in this sample, but does not establish that scalar aggregate orientation scores are insensitive audit instruments for LLM simulation in general. It does not rule out more granular channels of value sensitivity, which Diagnostics 2 and 3 examine directly.

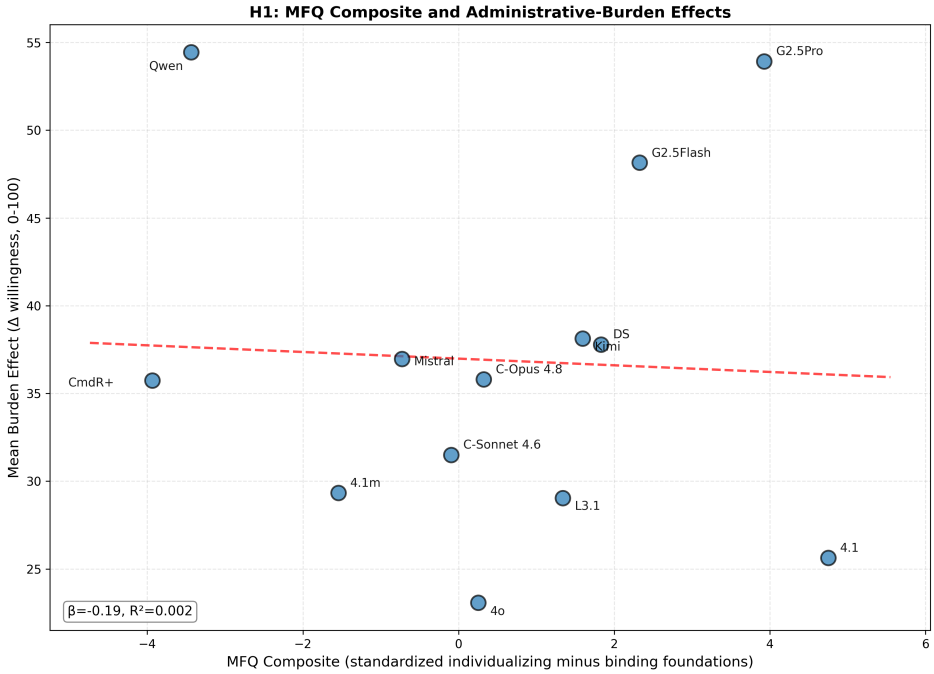


Figure 2: Moral-foundations composite and predicted administrative-burden effects

Notes: Each point represents one model in the thirteen-model sample. The horizontal axis reports the model's MFQ-30 composite (signed sum of standardized foundation scores: care + fairness − loyalty − authority − sanctity), and the vertical axis reports the model-level average burden effect, $\bar{\Delta}_m = \bar{W}_m^{low} - \bar{W}_m^{high}$. The fitted line shows the bivariate regression ($\hat{\beta} = -0.19$, 95% CI [−2.50, +2.12], $p = 0.875$, $R^2 = 0.002$). The MFQ composite does not significantly predict variation in simulated burden effects.

4.3.2 H2: Narrative Frame-Decomposition Analysis

H2 asks whether interpretive frames in model rationales co-vary with predicted burden effects, independently of any aggregate orientation score. It is designed as a frame–outcome association analysis, not as a causal pathway from MFQ to burden predictions.

Two of the five frames varied meaningfully in the high-burden rationales: access-barrier reasoning in 43.2% (after recoding to require an explicit link between procedural costs and reduced access, resource constraints, work flexibility, childcare, provider access, or socioeconomic vulnerability) and coercive-backlash in 7.7%. Collective-responsibility reasoning appeared in only 0.05% of rationales once the coding rule was corrected: the original rule matched the bare phrase "public health," which fired almost exclusively on "trust in public health (authorities)" (1,495 of the 2,019 rationales) and "public health center" (the vaccination site; 81 rationales) rather than on collective-responsibility language. The collective-responsibility rule was corrected on face-validity grounds before estimation of the reported H2 specifications; both the original and corrected coding rules are retained in the replication archive. After correction, only one rationale expressed a genuine collective-responsibility frame. Autonomy infringement and procedural legitimacy were near zero and are reported descriptively only.

Table 2 reports the estimates. Under the primary model-by-profile pair-fixed-effects specification, none of the retained rationale frames shows a detectable association with predicted burden effects (access barriers $\hat{b} = 0.12$, 95% CI $[-0.79, 1.03]$, BH-adjusted $p = 0.79$; coercive backlash 1.20, adjusted $p = 0.79$). Collective responsibility is excluded from the inferential H2 model because corrected coding yielded only one positive rationale, leaving within-pair variation in only one of 405 model-profile pairs; this provides insufficient support for meaningful estimation. A specification with separate model and profile fixed effects yields a positive access-barrier coefficient ($\hat{b} = 1.08$, 95% CI $[0.24, 1.92]$, adjusted $p = 0.024$), so that association does not survive absorbing pair-specific heterogeneity: it reflects cells in which a model systematically expresses access-barrier reasoning and also shows larger burden effects, rather than variation across repeated queries of the same cell.

Identification under the pair-fixed-effects specification is limited by construction. Only 7.8% of the variance in Δ occurs within model–profile pairs (182 of 405 pairs show no within-pair variation in Δ at all), and the frame indicators are constant across repetitions in 237 to 362 of the 405 pairs, so the coefficients are driven by the minority of pairs in which the expressed frame changes between repeated queries. The null estimates should therefore be interpreted as evidence that the previously observed access-barrier association is not robust to pair-specific heterogeneity, rather than as evidence that rationale framing is behaviorally irrelevant: with only 7.8% within-pair variation, the within-pair estimand is estimable but supported by sparse within-cell variation in this design.

Leave-one-out diagnostics behave accordingly: under the primary specification the access-barrier coefficient stays positive in 14 of 15 leave-one-model-out fits (-0.05 to $+0.36$) and in 25 of 27 leave-one-profile-out fits (-0.10 to $+0.32$), but is never statistically distinguishable from zero in either series, as is the coercive-backlash frame. H2 was not supported under the primary pair-fixed-effects specification.

4.3.3 H3: Within-Model Frame Manipulation

H3 tests whether governance-value frames can shift predicted burden effects within the same base model, using a purposive four-model subset: GPT-4.1, Llama 3.1 70B Instruct,

Table 2: Narrative frame–outcome associations under model-by-profile pair fixed effects

Frame	Prevalence	Pair-FE b	SE	95% CI	BH-adj. p	Two-way FE b
Access barriers	43.2%	0.12	0.46	[−0.79, 1.03]	0.79	1.08
Coercive backlash	7.7%	1.20	1.84	[−2.41, 4.81]	0.79	1.59
Collective responsibility	0.05%	—	—	—	—	—

Notes: The primary specification includes one fixed effect per model–profile pair (405 pairs) plus the retained rationale-frame indicators; coefficients are therefore identified exclusively from repetition-level changes in frame presence within the same model–profile pair, and standard errors are clustered at the model–profile level. $N = 2{,}019$. Collective responsibility is excluded from the primary specification because corrected coding yielded only one positive rationale, leaving within-pair variation in only one of 405 model-profile pairs. The two-way column reports a specification with separate model and profile fixed effects (additive heterogeneity only) and is shown for comparison. p-values are Benjamini–Hochberg adjusted across the retained frames ($q = 0.10$); both are non-significant under the primary specification. Prevalence is the share of high-burden rationales expressing the frame. Only 7.8% of the variance in the burden effect $\Delta = W^{low} - W^{high}$ occurs within model–profile pairs (182 of 405 pairs show no within-pair variation in Δ), and the frame indicators are constant across repetitions in 237 (access barriers) and 362 (coercive backlash) of the 405 pairs, so the pair-fixed-effects estimates rest on limited within-cell variation. Leave-one-out ranges for the primary specification and the full identification diagnostics are reported in Appendix H. The table reports frame–outcome associations rather than formal mediation: MFQ-to-frame paths and $a \times b$ products are not interpreted as causal mediation estimates because MFQ varies only at the model level and is not experimentally assigned.

Qwen3.7 Plus, and Mistral Large.

Figure 3 reports pooled frame effects against the neutral frame, with standard errors clustered on the 108 model–profile units (27 profiles × 4 models). The 432 model–profile–frame cells are design cells; estimation uses the 2,159 profile–repetition observations they contain. All three contrasts remain significant after Holm–Bonferroni adjustment: autonomy increases the predicted burden effect by 4.96 points (95% CI [3.57, 6.35]), collective obligation decreases it by 10.88 points ([-12.54, -9.21]), and equity/access increases it by 4.84 points ([3.45, 6.24]); adjusted $R^2 = 0.77$ ($N = 2{,}159$). Because the pooled estimates average heterogeneous model-level responses, the model-specific results are reported below and in the Online Supplement.

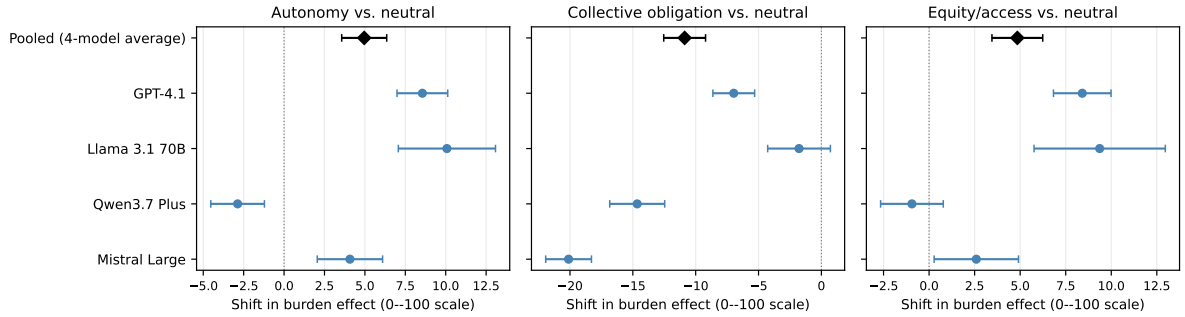
These are within-model–profile frame contrasts, not population-level causal effects: the observations are repeated queries of 27 constructed test cases, so the intervals reflect within-profile consistency of responses at temperature = 0 in this controlled set of constructed model–profile test cases rather than population-level precision. Effect sizes (4.96, 4.84, and 10.88 points pooled, up to 20.11 points model-specifically) are the primary quantities of interest, with uncertainty quantified by clustered standard errors.

Because each model-specific regression rests on 27 profile clusters, inference was rechecked with a wild cluster bootstrap-t and a finite-sample corrected covariance; no conclusion changes, including the marginal equity/access contrast for Mistral Large ($p = 0.027$ bootstrap, 0.030 corrected) and the autonomy reversal for Qwen3.7 Plus (-2.87, 95% CI $[-4.53, -1.22]$; $p = 0.002$). Model-specific point estimates followed the theoretically expected directions for all three contrasts in three of the four models; Qwen3.7 Plus reversed sign on autonomy and showed a small, statistically indistinguishable-from-zero negative equity/access contrast. Mistral Large—the lowest-MFQ model in the subset—shifts in the expected direction on all three frames. H3 thus provides the clearest evidence that an assigned prompt clause shifts simulated effects within a model (by up to 20 points

on a 0–100 scale, with negative collective-obligation point estimates in all four models, statistically distinguishable from zero in three), while the direction of that sensitivity is model-specific: a frame that amplifies burden sensitivity in one endpoint can attenuate it in another.

Results were essentially unchanged when model-by-profile pair fixed effects replaced separate model and profile fixed effects (pooled autonomy +4.96, collective obligation −10.88, equity/access +4.86; Appendix I.4), confirming that the assigned-frame contrasts are identified within the same model–profile cells.

Figure 3: Forest plot of within-model frame-manipulation effects



Notes: Coefficients represent the shift in predicted burden effect (points on a 0–100 willingness scale) relative to the neutral frame. Positive values indicate larger predicted burden effects. Pooled estimates use model and profile fixed effects with standard errors clustered at the model–profile level ($N = 2{,}159$, 108 clusters: 27 synthetic profiles × 4 base models, adjusted $R^2 = 0.77$). Model-specific estimates show consistent directions for the collective-obligation contrast, but the autonomy and equity/access contrasts differ in sign across base models. Model-specific coefficients and standard errors are reported in the Online Supplement.

4.4 Robustness Checks

Robustness checks leave every conclusion unchanged. The H1 null persists across dimension-specific and alternative constructions (each of the five MFQ foundations is fitted separately, none significant; Table S14; the MFQ composite slope stays between 2.5 and 3.4 in leave-one-profile-out fits, never reaching conventional significance); under the primary pair-fixed-effects specification no retained H2 frame is significant in any leave-one-model-out or leave-one-profile-out fit, and the access-barrier coefficient remains positive in 14 of 15 leave-one-model-out and 25 of 27 leave-one-profile-out fits while the other frames remain null; and all three H3 contrasts retain sign and significance in every leave-one-profile-out fit, with model-specific regressions showing that the pooled manipulation is not driven by any single base model. Detailed results are reported in the Online Supplement.

5 Discussion

This study develops and evaluates a three-level diagnostic framework for auditing normative-frame sensitivity in LLM-assisted policy simulation. Using administrative bur-

den in vaccination access as a controlled HSR testbed, we demonstrate that seemingly technical simulated policy-effect estimates depend on analyst-selected normative framing.

Summary of Diagnostic Results. Diagnostic 1 shows that a composite derived from the MFQ-30–derived probe does not predict cross-model variation. Diagnostic 2 does not: under model-by-profile pair fixed effects, none of the retained rationale frames predicts burden effects. The access-barrier association that appears when only additive model and profile heterogeneity is removed disappears once pair-specific heterogeneity is absorbed, and only 7.8% of the variance in the burden effect lies within model–profile pairs. Diagnostic 3 shows that assigned frames shift estimated effects—by 4.8 to 10.9 points pooled and up to 20.1 points model-specifically, with the autonomy contrast differing in sign across models.

Taken together, the three diagnostics establish that, in this controlled testbed, LLM-assisted policy simulation is prompt-frame sensitive. The same model, same synthetic profiles, and same policy scenario can produce materially different simulated policy-effect estimates depending on the governance-value frame activated by the prompt.

Contribution beyond prompt-sensitivity benchmarks. That an instruction to weight a factor moves an estimate is not, by itself, a surprising finding, and the contribution here does not rest on it. The paper's contribution is that this sensitivity propagates to the policy-effect estimand that HSR analysts consume—the simulated marginal effect of administrative burden—and that it can be systemically diagnosed through a three-level audit framework (model-level aggregate orientation, rationale-level co-expression, controlled fully-crossed intervention with randomized execution order). Three things follow from treating that movement as the estimand an analyst consumes. First, the shifts are expressed in the units the analyst reports—the simulated marginal effect of the burden condition—where pooled shifts of 4.8–10.9 points (up to 20.1 points model-specifically) are large relative to the precision of the simulated contrasts (pooled 95% CIs span roughly ± 1.4 points). Second, the contrast is identified within the same model–profile cell under fully-crossed frame conditions with randomized execution order, so it cannot be attributed to model or profile composition. Third, the direction is not uniform: the autonomy contrast takes different signs across endpoints, so no single model's frame response can stand in for "the LLM" response or be removed by one global adjustment. The deliverable is therefore the audit rather than the observation: an aggregate-orientation screen, a co-expression diagnostic that fails its strict test, and an assigned-frame sensitivity test reported with effect sizes.

The narrative analysis does not supply a robust within-cell signal. Under the primary pair-fixed-effects specification none of the retained frames is detectable, and the access-barrier association that appears under additive model and profile fixed effects does not survive pair-specific heterogeneity—a pattern consistent with cells whose baseline burden effect and rationale style are jointly model- and profile-specific. Because only 7.8% of the variance in Δ occurs within pairs, this is a failure of robustness rather than proof that rationale framing is behaviorally irrelevant. H2 is descriptive co-expression in any case rather than causal mediation, because willingness and rationales are generated together and associations may reflect post-hoc justification.

The strongest evidence comes from the within-model manipulation. In the four-model subset, the collective-obligation frame reduced simulated burden effects in all four models and the autonomy and equity/access frames increased them in three of four (Qwen3.7 Plus reversed sign on autonomy and showed no equity effect). What is manipulated is the assigned prompt clause, so the estimate is the causal effect of assignment to the

governance-frame prompt condition on the simulated effect—a swing of up to 20 points on a 0–100 scale—and the direction of that sensitivity is itself model-specific.

Relation to Prior Work. These results sit at the intersection of three literatures. *First, the synthetic-human literature:* large-scale replication work shows that LLMs often recover the direction of experimental effects while producing larger effect sizes and returning significant results where human studies found nulls [31], that LLM replacement of human participants can flatten and misportray identity groups [32], and that synthetic respondents are imperfect survey substitutes [33]. Our frame manipulation documents frame sensitivity as a potential source of variation in LLM-generated effect magnitudes: estimated magnitudes are partly a function of the frame the analyst activates, so variation in estimated magnitudes reflects both model behavior and analyst-selected framing. Recent work on hybrid estimation proposes combining biased candidate predictions under convex constraints to preserve consistency even when individual models are arbitrarily misspecified [34]; the present audit differs in aim—we diagnose fragility before estimation—but the two approaches are complementary. *Second, the measurement-instability literature:* if value measurement is sensitive to format and phrasing [18, 19], and if survey instruments do not behave as validated psychometric scales for models [35, 36], then a composite built from such items should not be expected to predict an external outcome—the H1 result provides little support for the predictive validity of this particular composite for cross-model burden effects in this endpoint sample. *Third, prompt-sensitivity work:* prior studies document format and phrasing sensitivity in opinion and scale measurements [19–22]; we extend it to simulated marginal effects of a health-policy condition and show that it survives a fully crossed within-model manipulation. Hybrid approaches that filter LLM-generated candidates before causal modeling reduce ATE bias substantially [37], yet they assume the underlying generation is stable across prompts; our frame-sensitivity results suggest this assumption requires explicit audit. This matters more for HSR than for NLP evaluation: in NLP, prompt sensitivity degrades measurement reliability, whereas in HSR a frame-dependent simulated effect enters policy deliberation directly. Because administrative burden already delays or forecloses care for the patients least able to absorb procedural costs [4–6], estimated burden effects that differ across assigned prompt frames raise distinct equity concerns.

The findings have three implications. First, audits should move beyond stated-opinion tests to examine how models simulate policy effects under different frames. Second, audits should measure specific dimensions and frame sensitivity rather than rely on coarse provenance labels. Third, prompt framing should be treated as part of the experimental design: simulations should not be reported as single neutral estimates when governance-value frames materially shift predicted effects. Analysts should report model identity, prompt framing, scenario wording, and robustness to alternative frames. Funders and journals could strengthen reporting standards by requiring prompt, model, and frame-sensitivity documentation for LLM-assisted policy simulations.

5.1 Limitations and Future Research

Six limitations qualify the findings. **(1) Model sample.** The cross-model analyses rest on thirteen commercially available endpoints spanning nine provider ecosystems (OpenAI, Anthropic, Google, Meta, Mistral, Cohere, Moonshot AI, DeepSeek, Qwen), which are a convenience sample of the models available at the time of data collection rather than a random or systematically stratified sample of model families, sizes, or training regimes.

The endpoints are not independent draws from a population of model families: three are from OpenAI, two from Anthropic, two from Google, and two from Meta, so provider-level variation is not separately identified from model-level variation. At N = 13, model-level tests have low power (the approximate 80% minimum detectable slope for a model-level composite is about 5.7 points), so H1 and the foundation-specific estimates should be read as descriptive diagnostics rather than high-powered population estimates. The null result for a composite derived from an MFQ-30–derived probe based on a human-validated questionnaire should be read accordingly: it is consistent with the absence of a linear association in this sample, but cannot prove that scalar aggregate value scores are insensitive audit instruments for LLM simulation. In addition, H2 and H3 reuse repeated queries of the same synthetic profiles; standard errors are clustered at the model–profile level, but the resulting intervals reflect within-profile consistency of LLM responses rather than population-level precision, which is why effect sizes and their uncertainty intervals rather than p-values are the primary quantities of interest. Finally, orientation was measured from a single canonical battery administration, so endpoint-level test-retest reliability is not quantified.

(2) Synthetic testbed. The twenty-seven standardized profiles and the vaccination-access scenario are controlled test cases, not a representative sample of U.S. vaccination programs or of the populations they serve. The income-verification and residency requirements in the high-burden condition are stylized procedural barriers typical of means-tested programs, not realistic requirements for vaccination access. The findings therefore speak to LLM-assisted policy simulation methodology—in particular, its sensitivity to normative framing—and not to the magnitude of administrative burden in real health service delivery or to the behavioral dynamics of vaccination uptake.

(3) Prompt and protocol dependence. Each diagnostic relies on a single analyst-chosen prompt per level, a fixed item order for the orientation battery, and temperature = 0. The reported magnitudes are therefore conditional on those design choices; we did not run a systematic sweep over prompt wording, role instructions, or item order, and the frame sensitivity documented here should be read as protocol-conditional rather than as a fixed property of the models. Applying the diagnostics therefore requires documenting the prompt and protocol under which the reported magnitudes were obtained. Part-specific score distributions for the MFQ-30 battery differed across most foundations (relevance-judgment differences of 0.17–0.36 points for CA, FA, LO, and AU), providing no evidence of a gross response-format collapse; however, because relevance and judgment sections contain different item content, semantic differentiation between the two tasks cannot be established from these distributions alone.

(4) H2 rationale coding. Frame indicators for H2 are produced by rule-based coding of model-generated rationales, audited by manual inspection of a random sample of 200 rationales (approximately 10% of the high-burden sample). That audit is a single-coder quality check rather than an independent validation: it was not blind to the algorithm's assignments and does not quantify false negatives, so no precision, recall, or agreement statistic is reported. The coding is coarse, the rationales are generated alongside the willingness score rather than elicited separately, and the resulting estimates are frame–outcome co-expression rather than causal mediation; observed associations may partly reflect post-hoc justification.

(5) H3 treatment bundling and placebo. The high-burden condition increases paperwork, documentation, waiting time, location friction, and scheduling delay simultaneously, so the estimated Δ reflects a multidimensional high-burden regime rather than

any single component. Component-specific and interaction effects would require factorial designs that vary burden dimensions independently. In addition, the neutral condition contains no matched-length placebo clause, whereas the three treatment frames each insert a clause instructing the model to "give special weight to" the respective governance value. Because the comparison is between a complete prompt with an inserted frame clause and a complete prompt without such a clause, H3 identifies the effect of assignment to each complete prompt-frame condition; it does not separately identify the normative semantic content of the clause from generic instruction salience or the addition of prompt text.

(6) Absence of human calibration. The framework contains no human ground truth. We can establish that simulated burden effects are sensitive to the assigned frame, but not whether the simulated magnitudes, or their frame sensitivity, correspond to those of human respondents. The diagnostics identify fragility, not validity of magnitude. Anchoring these simulations to externally calibrated human-subject data—for example, to trial-based or stated-preference benchmarks for the same policy scenario—is the natural next step, and would allow the frame-sensitivity results to be interpreted against a human reference point.

5.2 Conclusion

The central finding is that LLM-assisted policy simulation creates a new methodological problem for HSR: seemingly technical simulated policy-effect estimates may depend on analyst-selected normative framing. The three-level diagnostic framework demonstrates that (1) a composite derived from the MFQ-30–derived probe was not informative as a cross-model predictor of simulated burden effects, (2) naturally expressed rationale frames did not provide a robust within-pair diagnostic of simulated effect magnitude, and (3) experimentally assigned governance-value frames materially shift estimated policy effects.

For HSR researchers using LLMs for policy simulation, the implication is clear: report frame sensitivity rather than treating one prompt/model output as a neutral point estimate. Analysts should document model identity, prompt framing, scenario wording, and robustness to alternative frames. The goal is not to eliminate values from policy simulation, but to make their influence measurable, transparent, and contestable—so that the method can be used responsibly as an HSR tool.

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Online Supplement

A Framework for Auditing Value Sensitivity in LLM-Assisted Health Policy Simulation

Yichao Jin

October 2, 2026

This Online Supplement contains supporting materials for the main text. Appendices are presented in the order in which they are first cited.

A Data Quality and Simulation Diagnostics

Table S1 reports task-compliance diagnostics for the expanded fifteen-model simulation (4,050 raw responses; 2,025 possible model–profile–repetition pairs, of which 2,019 are analyzable because six high-burden responses failed structure checks). JSON validity, score-range compliance, and non-refusal rates were 100% for fourteen of fifteen models; Gemini 2.5 Flash had six malformed responses (97.8% compliance). The flip rate was 0% for every model, indicating that the high-burden condition moved mean predicted willingness in the expected direction in every profile, i.e., directional consistency rather than evidence that models understood the manipulation. Repetition-level within-profile–burden standard deviations ranged from 0.03 to 3.14 points on the 0–100 scale, indicating high within-model stability.

Table S1: Data quality and simulation diagnostics by model

Model	N	JSON%	Score%	NonRef%	Mean W	StabSD	DeltaSD	Flip%	ClassAcc%
Llama 3.1 70B	270	100.0	100.0	100.0	65.48	0.87	9.45	0.0	100.0
GPT-4o	270	100.0	100.0	100.0	64.80	0.97	9.08	0.0	100.0
GPT-4.1	270	100.0	100.0	100.0	61.85	0.69	8.27	0.0	100.0
Cohere Command R+	270	100.0	100.0	100.0	58.91	2.16	5.27	0.0	100.0
GPT-4.1 mini	270	100.0	100.0	100.0	58.81	0.94	4.45	0.0	100.0
Gemini 2.5 Flash	270	97.8	97.8	100.0	57.92	3.06	9.64	0.0	100.0
Mistral Large	270	100.0	100.0	100.0	57.52	0.40	6.07	0.0	100.0
DeepSeek-V3.2	270	100.0	100.0	100.0	56.86	1.24	5.28	0.0	100.0
Claude Haiku	270	100.0	100.0	100.0	56.59	0.03	8.23	0.0	100.0
Qwen3.7 Plus	270	100.0	100.0	100.0	56.10	2.23	7.43	0.0	100.0
Kimi K2.6	270	100.0	100.0	100.0	55.25	1.63	6.43	0.0	100.0
Llama 3.1 8B	270	100.0	100.0	100.0	52.76	3.14	7.23	0.0	100.0
Claude Opus 4.8	270	100.0	100.0	100.0	52.54	0.54	8.74	0.0	100.0
Gemini 2.5 Pro	270	100.0	100.0	100.0	50.48	3.03	8.00	0.0	100.0
Claude Sonnet 4.6	270	100.0	100.0	100.0	47.99	0.44	8.35	0.0	100.0

Notes: N reports the number of raw simulation responses per model. JSON% is the percentage of outputs parsed as valid JSON. Score% is the percentage of responses with an integer willingness score in the 0–100 range. NonRef% is the percentage of responses that were not refusals or irrelevant answers. Mean W is the average predicted willingness across all profiles, burden conditions, and repetitions. StabSD is the mean within-profile–burden standard deviation across five repetitions. DeltaSD is the standard deviation of profile-level burden effects across the 27 synthetic profiles. Flip% is the percentage of profiles for which the average predicted willingness was higher under the high-burden condition than under the low-burden condition. ClassAcc% is the burden-classification accuracy of the auxiliary comprehension diagnostic described in Appendix F: the share of low- and high-burden scenarios for which the model returned the correct burden label (54 scenarios per model, one call per scenario). All fifteen models returned the correct label for every scenario, so the flip rate documents directional consistency of the simulated responses while ClassAcc% documents recognition of the manipulation itself. The classification block was fielded on 17 September 2026 (UTC), separately from the canonical simulation runs collected on 16–17 September 2026.

B Model Versioning, API Endpoints, and Decoding Parameters

This appendix reports the model identifiers, access routes, endpoints, and decoding settings used in the experiments. The study includes fifteen LLM endpoints. An initial nine-endpoint design was expanded to fifteen, and all fifteen endpoints—the original nine and the six additional ones—were rerun under the same canonical protocol on 16–17 September 2026, so every simulation, frame-manipulation and orientation result reported here comes from a single administration; the auxiliary burden-classification diagnostic of Appendix F was fielded separately on 17 September 2026. Access routes differed by endpoint and are reported in Table S2: nine endpoints were reached through official provider APIs (OpenAI, Anthropic, Google, and DeepSeek), five through the OpenRouter gateway (Llama 3.1 70B Instruct, Llama 3.1 8B Instruct, Mistral Large, Kimi K2.6, and Cohere Command R+), and one through the Alibaba DashScope API (Qwen3.7 Plus). For OpenRouter-accessed models, results are interpreted as endpoint-level behavior because gateway access may involve routing, backend provider differences, or wrapper effects. Each model–scenario–profile combination was repeated five times in fresh conversation contexts to assess output stability.

Table S2: Model details, identifiers, endpoints, and decoding settings.

Model	Provider	Access route	Model identifier	Endpoint	Temp.	Decoding note
GPT-4.1	OpenAI	Official API	`gpt-4.1` returns `gpt-4.1-2025-04-14`	OpenAI Chat Completions API	0	Dated snapshot returned
GPT-4.1 mini	OpenAI	Official API	`gpt-4.1-mini` returns `gpt-4.1-mini-2025-04-14`	OpenAI Chat Completions API	0	Dated snapshot returned
GPT-4o	OpenAI	Official API	`gpt-4o` returns `gpt-4o-2024-08-06`	OpenAI Chat Completions API	0	Dated snapshot returned
Claude Sonnet 4.6	Anthropic	Official API	`claude-sonnet-4-6`	Anthropic Messages API	0	Near-deterministic; no seed
Claude Haiku 4.5	Anthropic	Official API	`claude-haiku-4-5` returns `claude-haiku-4-5-20251001`	Anthropic Messages API	0	Model versioning in response metadata
Gemini 2.5 Pro	Google	Official API	`gemini-2.5-pro`	Gemini API	0	Near-deterministic
Gemini 2.5 Flash	Google	Official API	`gemini-2.5-flash`	Gemini API	0	Near-deterministic
Llama 3.1 70B Instruct	Meta	OpenRouter	`meta-llama/llama-3.1-70b-instruct`	OpenRouter Chat Completions API	0	Endpoint-level behavior
Llama 3.1 8B Instruct	Meta	OpenRouter	`meta-llama/llama-3.1-8b-instruct`	OpenRouter Chat Completions API	0	Endpoint-level behavior
Mistral Large	Mistral	OpenRouter	`mistralai/mistral-large`	OpenRouter Chat Completions API	0	Endpoint-level behavior
DeepSeek-V3.2	DeepSeek	Official API	`deepseek-chat` returns `deepseek-flash`	DeepSeek API	0	Endpoint returns snapshot variant
Qwen3.7 Plus	Alibaba	DashScope API	`qwen3.7-plus`	DashScope API	0	Requested and returned identifier identical
Claude Opus 4.8	Anthropic	Official API	`claude-opus-4-8`	Anthropic Messages API	n/a	Temperature not accepted by this model; parameter omitted

Continued on next page

Table S2 – continued

Model	Provider	Access route	Model identifier	Endpoint	Temp.	Decoding note
Kimi K2.6	Moonshot AI	OpenRouter	`moonshotai/kimi-k2.6-20260420` returns `moonshotai/kimi-k2.6`	OpenRouter Chat Completions API	0	Dated snapshot requested, base version returned
Cohere Command R+	Cohere	OpenRouter	`cohere/command-r-plus-08-2024`	OpenRouter Chat Completions API	0	Explicit dated version used

Reproducibility Notes

All models were run with temperature set to zero where supported. Each API call used a fresh conversation context with no prior messages. The responses analyzed here were collected on 16–17 September 2026 (UTC) from model APIs only; no human participants were involved (see the Human Subjects statement in the main text). Replication code and derived data are released under the MIT License. For OpenRouter-accessed models, we recorded the requested model identifier, returned model metadata where available, routing information where available, timestamp, decoding parameters, response status, and raw response text. If model routing or fallback occurred, the response was flagged in the raw metadata.

Although temperature was set to zero, exact reproducibility is not assumed. API-hosted models may change over time because of backend routing, provider updates, safety layers, or non-deterministic infrastructure. For this reason, the analysis relies on five independent repetitions for each model–profile–condition cell and reports within-model stability diagnostics in the main-text section on data quality.

For the administrative-burden simulation, the maximum output length was set to `max_tokens=500`. For the moral-foundations battery (MFQ-30), the maximum output length was set to `max_tokens=2048`. Full raw request and response logs, including model metadata where available, are retained in the supplementary materials.

C Moral Foundations Questionnaire-30: Full Item Wording

This appendix reports the complete 30-item Moral Foundations Questionnaire (MFQ-30) [23] used to measure the five moral foundations described in Section 3.2, together with the two control items. For each model, items were presented in the fixed blocked order listed below (CA01–CA06, FA07–FA12, LO13–LO18, AU19–AU24, SA25–SA30, CT31–CT32). The instrument follows the original two-part administration: sixteen relevance items (Part 1: "When you decide whether something is right or wrong, to what extent are the following considerations relevant to your thinking?", scored 0–5) and fourteen agreement items (Part 2: "Please indicate your agreement with the following statements.", scored 0–5). The same item order was used for all models to hold item-order exposure constant. The scoring rule that turns the thirty ratings into the composite MFQ score is stated in the final subsection of this appendix.

Care/Harm (CA)

Relevance items. Rate each consideration from 0 (not at all relevant) to 5 (extremely relevant).

CA01 Whether or not someone suffered emotionally

CA02 Whether or not someone cared for someone weak or vulnerable

CA03 Whether or not someone was cruel

Judgment items. Rate each statement from 0 (strongly disagree) to 5 (strongly agree).

CA04 Compassion for those who are suffering is the most crucial virtue.

CA05 One of the worst things a person could do is hurt a defenseless animal.

CA06 It can never be right to kill a human being.

Fairness/Reciprocity (FA)

Relevance items. Rate each consideration from 0 (not at all relevant) to 5 (extremely relevant).

FA07 Whether or not some people were treated differently than others

FA08 Whether or not someone acted unfairly

FA09 Whether or not someone was denied his or her rights

Judgment items. Rate each statement from 0 (strongly disagree) to 5 (strongly agree).

FA10 When the government makes laws, the number one principle should be ensuring that everyone is treated fairly.

FA11 Justice is the most important requirement for a society.

FA12 I think it's morally wrong that rich children inherit a lot of money while poor children inherit nothing.

Ingroup/Loyalty (LO)

Relevance items. Rate each consideration from 0 (not at all relevant) to 5 (extremely relevant).

LO13 Whether or not someone's action showed love for his or her country

LO14 Whether or not someone did something to betray his or her group

LO15 Whether or not someone showed a lack of loyalty

Judgment items. Rate each statement from 0 (strongly disagree) to 5 (strongly agree).

LO16 I am proud of my country's history.

LO17 People should be loyal to their family members, even when they have done something wrong.

LO18 It is more important to be a team player than to express oneself.

Authority/Respect (AU)

Relevance items. Rate each consideration from 0 (not at all relevant) to 5 (extremely relevant).

AU19 Whether or not someone showed a lack of respect for authority

AU20 Whether or not someone conformed to the traditions of society

AU21 Whether or not an action caused chaos or disorder

Judgment items. Rate each statement from 0 (strongly disagree) to 5 (strongly agree).

AU22 Respect for authority is something all children need to learn.

AU23 Men and women each have different roles to play in society.

AU24 If I were a soldier and disagreed with my commanding officer's orders, I would obey anyway because that is my duty.

Purity/Sanctity (SA)

Relevance items. Rate each consideration from 0 (not at all relevant) to 5 (extremely relevant).

SA25 Whether or not someone violated standards of purity and decency

SA26 Whether or not someone did something disgusting

SA27 Whether or not someone acted in a way that God would approve of

Judgment items. Rate each statement from 0 (strongly disagree) to 5 (strongly agree).

SA28 People should not do things that are disgusting, even if no one is harmed.

SA29 I would call some acts wrong on the grounds that they are unnatural.

SA30 Chastity is an important and valuable virtue.

Control Items (excluded from scoring)

CT31 Whether or not someone was good at math

CT32 It is better to do good than to do bad.

Fixed Blocked Order (as presented to models)

1. CA01
2. CA02
3. CA03
4. CA04
5. CA05
6. CA06
7. FA07
8. FA08
9. FA09
10. FA10
11. FA11
12. FA12
13. LO13
14. LO14
15. LO15
16. LO16
17. LO17
18. LO18
19. AU19
20. AU20
21. AU21
22. AU22
23. AU23
24. AU24
25. SA25

26. SA26
27. SA27
28. SA28
29. SA29
30. SA30
31. CT31
32. CT32

Canonical Administration Protocol and Provenance

The battery was administered to all fifteen model endpoints under a single frozen protocol in the canonical rerun, so the fifteen MFQ values are measured on one instrument rather than on a mix of protocols. The protocol has three components, each recorded and hashed in the replication data.

System prompt (verbatim).

> You are completing a moral foundations questionnaire. Output only valid JSON.

SHA256 of the system prompt:
`1d0d8561a0960de7703af8e798af428a9fcaf42cc2d39f9702aad33af132bea2`

User prompt. The rendered user prompt opens with the two-part instruction (Part 1 relevance on 0–5 scale, Part 2 judgment on 0–5 scale), then lists the 32 items in the fixed order above as `R<order>. [<item_id>] <item text>` and `J<order>. [<item_id>] <item text>`, and closes with a request for a single JSON object mapping item ids to scores, e.g. `{"CA01": 4, "FA02": 3, ...}`.

Output schema and parsing. Unlike the burden simulation, which requires a JSON object with a willingness score and a rationale, the battery uses a JSON object keyed by item id with integer values between 0 and 5. A response counts as fully parsed only if all 32 item ids are recovered with scores in range. All fifteen endpoints returned fully parsable responses in the canonical run; the two control items are excluded from scoring.

Provenance fields. Each battery call is logged with the model identifier, provider, access route, the SHA256 of the system prompt and of the rendered user prompt, the raw response text, the parsed score vector, and a UTC timestamp; the protocol lock that fixes these strings is committed alongside the runner.

MFQ Scoring

The composite is computed in three steps, all from the thirty item ratings and all on the same sample of model endpoints.

1. Dimension scores. Each foundation score is the unweighted mean of its six items (three relevance, three judgment):

$$\mathrm{MFQ}_m^{(d)} = \frac{1}{6}\sum_{i=1}^{6} x_{m,i}^{(d)},$$

where $x_{m,i}^{(d)}$ is model m's rating on item i of foundation d, on the 0–5 scale. **2. Standardisation.** Each foundation score is standardised across endpoints:

$$Z_m^{(d)} = \frac{\mathrm{MFQ}_m^{(d)} - \bar{x}^{(d)}}{s^{(d)}}.$$

3. Composite. Following the standard individualizing-minus-binding convention in the moral-foundations literature [23], the composite is computed as the unweighted sum of the five standardized foundation scores:

$$\mathrm{MFQ}_m = Z_m^{(\mathrm{CA})} + Z_m^{(\mathrm{FA})} - Z_m^{(\mathrm{LO})} - Z_m^{(\mathrm{AU})} - Z_m^{(\mathrm{SA})},$$

where CA = Care/Harm, FA = Fairness/Reciprocity, LO = Ingroup/Loyalty, AU = Authority/Respect, and SA = Purity/Sanctity. Care and fairness (individualizing foundations) enter positively; loyalty, authority, and sanctity (binding foundations) enter negatively, encoding the standard direction in which each foundation is expected to relate to prosocial and policy judgments. The composite is *not* rescaled after aggregation, so its standard deviation is not exactly one; in the fifteen-endpoint sample its standard deviation is 1.17.

D Standardized Synthetic Profiles: Controlled Test Cases

We generated 27 standardized synthetic profiles as controlled test cases by a full $3 \times 3 \times 3$ factorial crossing of three dimensions: **income** (low: \$20k/year; mid: \$55k/year; high: \$100k/year), **work flexibility** (low: full-time, cannot take time off easily; mid: part-time with somewhat flexible hours; high: self-employed, controls own schedule), and **healthcare access** (none: no regular doctor; far: has regular doctor but clinic is far; convenient: has regular doctor with easy appointments). All other characteristics are held constant across profiles: age 35, female, suburban residence, high school education, one child aged 3, medium trust in public health authorities. These profiles are experimental stimuli designed to isolate how models interpret the same administrative-burden scenarios under different model and frame conditions; they are not intended to represent a human population.

Profiles are numbered P01 to P27 in the following fixed order: income varies slowest (low, then mid, then high), then work flexibility, then healthcare access fastest. Table S3 lists each profile's identifier and full natural-language description as presented to models.

Table S3: 27 standardized synthetic profiles (controlled test cases).

ID	Natural-language description
P01	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P02	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P03	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P04	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P05	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.

Table S3 – continued

ID	Natural-language description
P06	A 35-year-old woman living in a suburban area. Her household income is about $20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P07	A 35-year-old woman living in a suburban area. Her household income is about $20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P08	A 35-year-old woman living in a suburban area. Her household income is about $20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P09	A 35-year-old woman living in a suburban area. Her household income is about $20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P10	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P11	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P12	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P13	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P14	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.

Table S3 – continued

ID	Natural-language description
P15	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P16	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P17	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P18	A 35-year-old woman living in a suburban area. Her household income is about $55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P19	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P20	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P21	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P22	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.

Table S3 – continued

ID	Natural-language description
P23	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P24	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P25	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P26	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P27	A 35-year-old woman living in a suburban area. Her household income is about $100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.

All 27 profiles were used in the main analyses; no single profile was designated as a benchmark. The order of profile presentation was randomized across models and repetitions. Robustness checks re-estimate the main models excluding each synthetic profile one at a time; results are reported in Appendix J.

E Capability and Task-Diagnostic Details

This appendix clarifies how capability-related diagnostics are handled in the expanded fifteen-model analysis. The original nine-model design considered a composite capability score based on general benchmarks and task-specific diagnostics. In the expanded sample, however, comparable general benchmark scores are not uniformly available for all fifteen endpoints, especially for the endpoints added when the sample was expanded from nine to fifteen, which span additional vendors and access routes. Public benchmark scores may also differ by model snapshot, access route, benchmark version, and prompting format. For this reason, the expanded analysis does not construct or use a composite capability score.

Task-specific diagnostics are reported in Table S1. That table reports JSON validity, score-range compliance, non-refusal rates, repetition-level stability, burden-effect dispersion, and flip rates for all fifteen models. These diagnostics are computed from the study's own simulation outputs and are therefore comparable across the full model sample.

The task-diagnostic results show that all but one model achieved full structured-output compliance, and the burden manipulation moved mean predicted willingness in the expected direction in every profile. Because these diagnostics are already reported in Table S1, they are not repeated here. The substantive analyses therefore focus on measured moral foundations, predicted administrative-burden effects, narrative frames, and within-model frame responsiveness rather than on capability-adjusted regressions.

Evaluation Configuration

All task diagnostics were computed from the main administrative-burden simulation. Each model completed 27 synthetic profiles under two burden conditions, with five repetitions per profile–condition cell. For Gemini 2.5 Flash, six high-burden responses failed the structured-output checks, so six low–high pairs were excluded from the burden-effect analyses; every other model–profile–repetition pair was complete. All API calls used fresh conversation contexts. Temperature was set to zero where supported. Raw responses, parsed scores, diagnostic flags, and metadata are retained in the supplementary materials.

F Administrative-Burden Simulation Prompt Templates

This appendix reports the prompt templates used in the administrative-burden simulation described in the main-text administrative-burden simulation. Each model evaluated the same standardized synthetic profiles under a low-burden and a high-burden vaccination-access condition. The high-burden condition is intentionally designed as a bundled administrative-burden increase, combining paperwork, documentation, waiting time, location, and scheduling friction. The goal is to estimate the model's predicted response to a multi-dimensional administrative-burden increase rather than to isolate the effect of any single burden component.

System prompt

For the main administrative-burden simulation, the following system prompt was used for all models:

> You are estimating how likely the person described below would be to get vaccinated under the specified access conditions. Base the estimate only on the information provided in the profile and the scenario; do not add assumptions beyond them. Respond ONLY with a single JSON object and nothing else. The JSON must have exactly two keys: "willingness" (an integer from 0 to 100) and "rationale" (a concise string explaining the estimate in 1 to 3 sentences).

SHA256 of the system prompt:
`380d779e199673cdeaf808731aad01414b60e71edba7c23a57602241ff6dcffe`

User prompt template

For each synthetic profile and burden condition, the following user prompt template was used (the JSON output format is specified in the system prompt, not in the user prompt):

> Person profile:
>
> `{PROFILE_TEXT}`
>
> Scenario:
>
> `{SCENARIO_TEXT}`
>
> What is this person's willingness (0–100) to proceed with vaccination given this registration scenario?

SHA256 of the template string:
`1b98f920c7a2a1932b741233e800f99a62e1de84a1a172cd2038f1ad3c6970e4`

The rendered user prompt therefore consists of the profile text and scenario text documented in this appendix, inserted into the template above. The two burden scenarios are fixed strings; their SHA256 hashes are:

low burden:
`7307a8b8f499ebc0897c6fa0436fb8eafb3e77efeebf21ce91d8471989ec9b01`
high burden:
`c0bc058f951e4294842a4173f4cc2ac1e0475f1ff84d9438151cc9bd42967b52`

The 27 profile strings are the natural-language descriptions listed in Appendix D.

Low-burden scenario

The low-burden condition used the following scenario text:

> The person can schedule the vaccination appointment online. The appointment is available at a nearby pharmacy with extended evening and weekend hours. No documentation is required beyond a standard identification card. The appointment can be confirmed immediately, and the estimated total time required, including check-in and vaccination, is about 15 minutes.

High-burden scenario

The high-burden condition used the following scenario text:

> The person must register in person at a designated public health center. The person must bring income verification and proof of residency. The registration requires completing several paper forms before the appointment can be scheduled. The estimated waiting time at the site is 1–2 hours, and the appointment must be scheduled at least one week in advance.

Pairing of low- and high-burden conditions

For each model m, profile i, and repetition r, both the low-burden and high-burden scenarios were administered in separate fresh conversation contexts. The marginal burden effect was computed as:

$$\Delta_{mir} = W_{mir}^{\text{low}} - W_{mir}^{\text{high}}.$$

Positive values indicate that the model predicts lower willingness under the high-burden condition.

Burden-level comprehension diagnostic

Distinguishing comprehension from directional consistency requires a diagnostic that asks models directly about the manipulation. We therefore fielded an auxiliary burden-classification prompt with the same profile and scenario text, administered outside the frozen canonical run (17 September 2026 UTC; the canonical simulation, frame-manipulation, and orientation data were collected on 16–17 September 2026):

> **System prompt (verbatim):** You are answering a single factual question about a vaccination access scenario. Answer with one word only.
>
> Individual profile:
>
> `{PROFILE_TEXT}`

Vaccination access scenario:

`{SCENARIO_TEXT}`

Question:

Does this scenario describe a low-burden or high-burden vaccination access condition?

Return only one word: `low` or `high`.

SHA256 of the classification system prompt:
`4bf68c6c4da0a644e4c1ddb43bddbb0d9fa549ed7a978fccad5d8e2fa8107a04`
SHA256 of the execution plan (`canonical/plans/plan_comp.csv`):
`9677719485949d6959c0a5fbe5e4dfd93cc608370497e05865e57a1378c96c22`

The plan covers every combination of the fifteen endpoints, the 27 standardized profiles, and the two burden conditions (810 calls; order randomised per model from the protocol seed, as in the main blocks). A response was scored as correct if the returned label matched the burden condition assigned to the scenario that was shown, and as malformed if it contained neither token.

Results. All 810 responses returned a usable label and none was malformed. Accuracy was 100% for every model (54 of 54 scenarios each; exact Clopper–Pearson 95% CI [93.4, 100.0]) and 100% pooled (810 of 810; exact 95% CI [99.5, 100.0]). The models therefore recognised the burden manipulation itself, and not merely its directional signature in the willingness responses. Two qualifications apply. First, the diagnostic establishes recognition of the assigned condition, not the magnitude or realism of the simulated burden effect. Second, it was fielded in a separate administration from the reported simulation runs, so it documents comprehension of the same scenario texts rather than of the exact queries that produced the main results.

G Narrative Coding Codebook

This appendix reports the algorithmic coding protocol used to classify high-burden rationales into five interpretive frames. The unit of coding is the rationale (one to three sentences) associated with each model–profile–repetition observation under the high-burden condition. Each frame is coded as a binary indicator using regular-expression pattern matching. Frames are not mutually exclusive.

G.1 Algorithmic Coding Procedure

Rationales are coded using a rule-based algorithm that applies predefined keyword patterns (regular expressions) to each rationale text. The algorithm operates as follows:

1. **Text preprocessing:** Rationale text is converted to lowercase.
2. **Pattern matching:** For each of the five frames, the algorithm searches for predefined regular expression patterns (word boundaries applied where appropriate).
3. **Binary assignment:** A frame is coded as present (1) if at least one pattern matches; absent (0) otherwise.
4. **Multiple frames:** Because frames are not mutually exclusive, a single rationale may be coded as positive for multiple frames.

G.2 Pattern Dictionary

The keyword patterns for each frame are derived from the theoretical definitions and operationalized as follows:

Autonomy infringement: `individual choice`, `personal choice`, `personal freedom`, `autonomy`, `privacy`, `consent`, `personal discretion`, `right to choose`, `individual liberty`, `procedural dignity`, and phrases indicating limits on personal autonomy.

Procedural legitimacy: `justified`, `safeguard`, `reasonable verification`, `necessary procedure`, `eligibility`, `fraud prevention`, `legitimate public need`, `due process`, `accountability`, `oversight`, and phrases indicating procedural justification.

Collective responsibility: `civic duty`, `protect others`, `communal benefit`, `vulnerable group`, `collective welfare`, `common good`, `herd immunity`, and phrases emphasizing community benefit. The earlier `public health` pattern was removed because it matched "trust in public health (authorities)" and "public health center" (the vaccination site), which express trust in authorities or location rather than collective responsibility.

Access barriers: `cannot afford`, `miss work`, `lose income`, `childcare`, `no regular doctor`, `lack of access`, `low income`, `unable to comply`, `disadvantage`, `single parent`, `hourly worker`, and phrases linking burdens to resource constraints.

Coercive backlash: `resentment`, `distrust`, `resistance`, `backlash`, `reduced legitimacy`, `mandate`, `penalty`, `enforcement`, `backfire`, `coercion`, and phrases indicating reactance to mandates.

The full regular expression patterns are available in the replication code at `https://github.com/yichao2022/value-sensitivity-llm-audit`, which is released under the MIT License (Copyright 2026 Yichao Jin); reuse and modification are permitted with attribution, and model outputs remain subject to the providers' terms of service.

G.3 Manual Coding Audit

The algorithmic coding was audited, not independently validated. A stratified random sample of 200 rationales (approximately 10% of the 2,019 high-burden rationales, allocated proportionally across the fifteen endpoints) was inspected by the author. The audit procedure was:

1. **Scope:** the auditor reviewed each rationale together with the algorithm's frame assignments, without access to model identity or predicted outcomes. The auditor was therefore *not* blind to the algorithm's labels, and this is a single-coder quality check rather than an independent second coding.

2. **Definition check:** each positive algorithmic assignment was checked against the frame definitions in Table S4.

3. **Focus areas:** special attention was given to access-barrier coding, where generic mentions of paperwork or waiting time are not sufficient for a positive code under the algorithmic rules.

4. **Discrepancy review:** any rationale in which a pattern match appeared to violate a frame definition was flagged for review.

The manual audit was conducted after the collective-responsibility pattern correction. The audit found that pattern assignments reflected the codebook definitions rather than superficial keyword matches, and no systematic misclassification pattern was identified that would require pattern revision.

Two limits follow from the audit design. First, because the auditor saw the algorithm's assignments and concentrated on positive ones, the audit cannot bound false negatives: a frame the algorithm missed may simply not have been noticed, and the audit yields no precision, recall, F1, or agreement statistic. Second, single-coder inspection is not independent of the algorithm it checks. The main text therefore describes the frame indicators as descriptive evidence of rationale–outcome co-expression rather than as validated measurements.

A blinded re-coding sheet is archived in the replication materials for a human-versus-rule validation of the coding: `outputs/h2_gold_coding_sheet.csv` contains the 200 sampled rationale texts with the frame labels, model identity, profile, repetition, and outcome withheld into a separate key file, and the sampling is reproducible from the canonical data with seed 20260917 (`prepare_h2_gold_sample.py`). Coding the sheet and running `score_h2_gold_coding.py` yields per-frame precision, recall, F1, exact agreement, and Cohen's κ against the rule-based assignments. Those statistics are not reported here because the sheet was not coded in this study; the audit above is what was actually performed.

G.4 Frame Prevalence and Exclusions

Table S4: Narrative coding codebook

Frame	Code as 1 when rationale...	Do not code as 1 when rationale...
Autonomy infringement	Frames procedures as threats to individual choice, autonomy, privacy, personal freedom, consent, or procedural dignity.	Mentions inconvenience, waiting, forms, or documentation without linking them to autonomy, privacy, dignity, or choice.
Procedural legitimacy	Frames administrative requirements as justified, reasonable, legitimate, or necessary safeguards for verification, program integrity, safety, or orderly administration.	Mentions documentation or registration without justifying it, or treats procedures only as obstacles.
Collective responsibility	Emphasizes civic duty, public health responsibility, protection of others, community benefit, or willingness to tolerate inconvenience for collective welfare.	Mentions only individual health benefits or convenience without reference to public health, community, or others.
Access barriers	Explicitly links paperwork, waiting time, documentation, travel, scheduling, or in-person registration to reduced access, inability to comply, unequal burden, low income, work constraints, childcare, lack of regular provider access, or other vulnerability.	Merely restates that the scenario involves forms, documentation, waiting time, travel, or delay without linking these requirements to access, resources, or vulnerability.
Coercive backlash	Mentions resentment, distrust, resistance, backlash, reactance, reduced legitimacy, or negative reaction to mandates, penalties, enforcement, or perceived coercion.	Describes practical inconvenience or access difficulty without resistance, distrust, resentment, or coercion.

The access-barrier definition was intentionally restricted to avoid coding simple restatements of the high-burden prompt as a substantive frame. For example, "the process requires forms and a long wait" is not coded as access barriers unless the rationale links those requirements to reduced access or profile-specific constraints such as work flexibility, childcare, income, transportation, or lack of regular care.

Frames with little or no empirical variation are reported descriptively but excluded from the main frame-adjusted model. In the present analysis, autonomy infringement and procedural legitimacy each appeared in one of the 2,019 high-burden rationales (0.05%

each), and collective responsibility appeared in a single rationale (0.05%) after the coding rule was corrected (the original bare "public health" pattern, which matched "trust in public health" and "public health center," is removed). None of these three frames has usable variation. The main H2 model therefore retains access barriers and coercive backlash.

H H2 Specification Comparison and Identification Diagnostics

The main-text H2 analysis uses one fixed effect per model–profile pair, so its frame coefficients are identified only from repetition-level variation within a pair. This appendix reports the secondary specification with separate model and profile fixed effects, the identification diagnostics that follow from absorbing the pair-specific baseline, and leave-one-out variants of the primary specification.

Table S5: H2 frame–outcome estimates under the primary and secondary specifications

Frame	Prevalence	Pair-FE b	SE	95% CI	BH-adj. p	Two-way FE b
Access barriers	43.2%	0.13	0.47	[−0.79, 1.04]	0.86	1.08
Collective responsibility	0.05%	—	—	—	—	—
Coercive backlash	7.7%	1.21	1.82	[−2.35, 4.77]	0.86	1.59

Notes: Pair-FE is the primary specification of the main text: one fixed effect per model–profile pair (405 pairs) plus the retained rationale-frame indicators, with standard errors clustered at the model–profile level. Collective responsibility is dropped from the pair-FE column because the corrected coding leaves only one positive rationale (0.05%), giving the indicator no within-pair variation. Two-way FE replaces the pair indicators with separate model and profile fixed effects, absorbing additive heterogeneity only; its unadjusted p-values are 0.012 (access barriers) and 0.174 (coercive backlash), and its Benjamini–Hochberg adjusted values (across the two estimable frames) are 0.024 and 0.174. All p-values in the pair-FE column are Benjamini–Hochberg adjusted across the two retained frames ($q = 0.10$; collective responsibility is not estimable under this specification). $N = 2{,}019$ in both specifications. Positive coefficients indicate larger predicted burden effects.

Identification diagnostics

Table S6: Within-pair variation available to the pair-fixed-effects specification

Variable	Within-pair variance share	Pairs with no within-pair variation
Access barriers	0.329	237 of 405
Collective responsibility	0.800	404 of 405
Coercive backlash	0.310	362 of 405
Burden effect Δ	0.078	182 of 405

Notes: Each pair contains five repetitions. "Within-pair share of variance" is the fraction of a variable's total variance that remains after subtracting pair means, i.e. the variation the pair-fixed-effects estimator can use. Frame indicators that are constant within a pair contribute nothing to identification, so the frame coefficients rest on the pairs in which the expressed rationale changes between repeated queries of the same model–profile cell.

Two features of the design drive the results. First, the burden effect Δ is largely a property of the cell rather than of the individual query: only 7.8% of its variance lies within model–profile pairs, and in 182 of 405 pairs the five repetitions do not vary at all. Second, the coded rationale frames are also mostly stable within a pair, so identification comes from the minority of pairs in which the frame switches on or off across repetitions;

in the remaining pairs the frame indicator is absorbed by the pair effect. The pair-fixed-effects null in the main text should be read against these limits: it shows that the access-barrier association is not robust to pair-specific heterogeneity, and that the strict within-cell estimand is weakly identified in a five-repetition design.

Leave-one-out variants under the primary specification

Table S7: Leave-one-out ranges under the primary (pair-fixed-effects) specification

Frame	LOMO range	LOMO sig.	LOPO range	LOPO sig.
Access barriers	−0.03 to +0.37	0/15	−0.09 to +0.32	0/27
Coercive backlash	+0.69 to +1.58	0/15	+0.37 to +2.33	0/27

Notes: LOMO = leave-one-model-out (fifteen fits, each excluding one model); LOPO = leave-one-profile-out (twenty-seven fits, each excluding one profile). No frame coefficient is statistically distinguishable from zero in any leave-one-out fit under the primary specification. Sign counts: access barriers positive in 14 of 15 LOMO and 25 of 27 LOPO fits; coercive backlash positive in 15 of 15 and 27 of 27.

I Frame-Manipulation Prompts and H3 Robustness

This appendix reports the prompt templates and model-specific robustness checks for the within-model frame-manipulation experiment described in the main-text section on within-model frame manipulation and analyzed in the main-text H3 results. The purpose of the experiment is to test whether governance-value frames shift predicted administrative-burden effects while holding the base model, synthetic profiles, burden scenarios, output format, and decoding settings constant.

I.1 Frame-Manipulation Prompt Templates

The frame-manipulation experiment used four system-prompt frames. Each frame is built from the same two fixed sentences: an opening sentence and a closing instruction block, with the frame clause inserted between them. The neutral frame contains no frame clause, so the neutral system prompt is byte-identical to the baseline prompt reported in Appendix F:

`SHA256 380d779e199673cdeaf808731aad01414b60e71edba7c23a57602241ff6dcffe`

The frame contrast is therefore measured against the true baseline condition. The user prompt, synthetic profile, low- and high-burden scenario texts, and JSON output format were identical to those used in the main administrative-burden simulation reported in Appendix F.

> **Opening sentence:** You are estimating how likely the person described below would be to get vaccinated under the specified access conditions.
>
> **Closing instruction block:** Base the estimate only on the information provided in the profile and the scenario; do not add assumptions beyond them. Respond ONLY with a single JSON object and nothing else. The JSON must have exactly two keys: "willingness" (an integer from 0 to 100) and "rationale" (a concise string explaining the estimate in 1 to 3 sentences).

The four frame conditions differ only in the inserted clause:

- **Neutral frame:** no clause inserted.
  `SHA256 380d779e199673cdeaf808731aad01414b60e71edba7c23a57602241ff6dcffe`
- **Autonomy frame:** "When forming the estimate, give special weight to individual choice, privacy, personal autonomy, and procedural dignity."
  `SHA256 8b7d948eeee661603aa9898815ee3a01a058f43d4cc878cbc077e6cb5a5df243`
- **Collective-obligation frame:** "When forming the estimate, give special weight to civic responsibility, public health coordination, and willingness to tolerate reasonable inconvenience for collective protection."
  `SHA256 fe7512cde053405edf59cbaaf6d1402f68d69e7c478984eee3d0fbc3a49a5ecb`
- **Equity/access frame:** "When forming the estimate, give special weight to unequal procedural costs, access barriers, work constraints, childcare constraints, and burdens faced by disadvantaged groups."
  `SHA256 cba0156ababefd86611eb267ebf9ccddb1d60fe5a8cecf4e289871006042844a`

Randomization and archived execution plans. Execution order was randomized within each model by a generator seeded from the protocol seed (20260916, recorded in the protocol lock together with the prompt texts and decoding settings): in the baseline simulation the profile–scenario–repetition cells were shuffled, and in the frame manipulation the four frames were permuted within each profile–repetition cell, with the low/high order inside each pair randomized as well. The plans were generated once, frozen before data collection, and are archived with the replication materials:

Baseline simulation plan (`canonical/plans/plan_sim.csv`):
`097667f0bfe2b409fb9e37f6ceef4a6d05e3828eb25f55f7736de2005d2c350c`
Frame-manipulation plan (`canonical/plans/plan_h3.csv`):
`2da307521db1f8b578d1ebc87e77d85be372a1bdca5a10c5416069d958174133`

The runner consumes each plan in the recorded order, so the frame sequence actually administered to a given model is verifiable from the archived file rather than from the runner code alone. Rebuilding the plan from the current code reproduces the same set of cells but not necessarily the same row order; the archived files, not the builder, are the record of the executed sequence. The orientation battery, by contrast, was administered in a single fixed item order to all fifteen endpoints.

For each model m, frame condition c, profile i, and repetition r, the model evaluated both the low-burden and high-burden vaccination-access scenarios in fresh conversation contexts. The frame-specific burden effect was computed as:

$$\Delta_{mcir} = W^{low}_{mcir} - W^{high}_{mcir}.$$

Positive values indicate a larger predicted reduction in vaccination willingness under the high-burden condition.

I.2 Mean Burden Effects by Model and Frame

Table S8 reports mean burden effects by model and frame condition for the four-model H3 subset. The descriptive means show that the collective-obligation frame lowers simulated burden effects in all four models, and that the autonomy frame increases simulated burden effects in three of the four models. The remaining contrasts are model-specific rather than uniform: for Qwen3.7 Plus the autonomy frame lowers the mean burden effect (53.71 vs. 56.59 under the neutral frame), and the equity/access frame shifts that model only slightly (55.64 vs. 56.59 under neutral). Equity/access is instead sizeable and precisely estimated for Llama 3.1 70B Instruct (36.42 vs. 27.11 under neutral; regression coefficient +9.36, $p < 0.001$), where it is smaller than the autonomy shift but not negligible. The absolute level of the burden effect also differs substantially across models under the neutral frame (25.59 for GPT-4.1 to 56.59 for Qwen3.7 Plus), so frame effects are interpreted relative to each model's own neutral baseline.

Mistral Large is a particularly informative robustness case. It has the lowest MFQ composite score in the expanded fifteen-model sample, and all three frame manipulations shift its simulated burden effects in the expected directions. This suggests that frame sensitivity is not limited to models with high baseline MFQ or high baseline burden sensitivity.

Table S8: Mean burden effects by model and frame

Model	Frame	Mean Δ	SD	N
GPT-4.1	Neutral	25.59	8.42	135
GPT-4.1	Autonomy	34.15	6.84	135
GPT-4.1	Collective obligation	18.62	6.22	135
GPT-4.1	Equity/access	34.00	9.46	135
Llama 3.1 70B Instruct	Neutral	27.11	9.69	135
Llama 3.1 70B Instruct	Autonomy	37.19	7.98	135
Llama 3.1 70B Instruct	Collective obligation	25.33	10.42	135
Llama 3.1 70B Instruct	Equity/access	36.42	12.90	134
Qwen3.7 Plus	Neutral	56.59	7.64	135
Qwen3.7 Plus	Autonomy	53.71	6.54	135
Qwen3.7 Plus	Collective obligation	41.93	11.34	135
Qwen3.7 Plus	Equity/access	55.64	6.91	135
Mistral Large	Neutral	37.04	6.44	135
Mistral Large	Autonomy	41.11	5.38	135
Mistral Large	Collective obligation	16.93	7.53	135
Mistral Large	Equity/access	39.63	6.95	135

Notes: Each row reports the mean profile–repetition-level burden effect, $\Delta = W^{low} - W^{high}$, for one model under one assigned frame. $N = 135$ for each model–frame cell, corresponding to 27 profiles and five repetitions, except where a response failed the structured-output checks: Llama 3.1 70B Instruct equity/access ($N = 134$). Positive values indicate larger predicted reductions in willingness under the high-burden condition.

I.3 Small-Sample Inference for the Model-Specific Estimates

Each model-specific regression rests on 27 profile clusters, so we checked the model-level conclusions against two small-sample procedures: a cluster covariance with the standard finite-sample correction ($G/(G-1) \cdot (N-1)/(N-K)$) and a wild cluster bootstrap-t with Rademacher weights, the null imposed, and 1,999 replications. Table S9 reports the resulting p-values next to the asymptotic cluster-robust ones. The three inferences agree: the marginal equity/access contrast for Mistral Large remains significant (2.59, $p =$ 0.023 asymptotic, 0.030 corrected, 0.027 bootstrap), the negative autonomy contrast for Qwen3.7 Plus remains significant (-2.87, $p = 0.001$–0.002), and the collective-obligation contrast for Llama 3.1 70B Instruct and the equity/access contrast for Qwen3.7 Plus stay statistically indistinguishable from zero under every procedure.

Table S9: Small-sample inference for model-specific H3 contrasts

Model	Contrast	Coefficient	p (asymptotic)	p (corrected)	p (wild bootstrap)
GPT-4.1	Autonomy	+8.56	< 0.001	< 0.001	< 0.001
GPT-4.1	Collective obligation	−6.97	< 0.001	< 0.001	< 0.001
GPT-4.1	Equity/access	+8.41	< 0.001	< 0.001	< 0.001
Llama 3.1 70B Instruct	Autonomy	+10.07	< 0.001	< 0.001	< 0.001
Llama 3.1 70B Instruct	Collective obligation	−1.78	0.137	0.155	0.125
Llama 3.1 70B Instruct	Equity/access	+9.36	< 0.001	< 0.001	< 0.001
Qwen3.7 Plus	Autonomy	−2.87	< 0.001	0.001	0.002
Qwen3.7 Plus	Collective obligation	−14.65	< 0.001	< 0.001	< 0.001
Qwen3.7 Plus	Equity/access	−0.94	0.250	0.271	0.265
Mistral Large	Autonomy	+4.07	< 0.001	< 0.001	< 0.001
Mistral Large	Collective obligation	−20.11	< 0.001	< 0.001	< 0.001
Mistral Large	Equity/access	+2.59	0.023	0.030	0.027

Notes: Each model-specific regression includes profile fixed effects and frame indicators, with standard errors clustered by profile (27 clusters per model). Asymptotic p-values use the cluster-robust covariance without small-sample correction; corrected p-values apply $G/(G-1) \cdot (N-1)/(N-K)$; wild bootstrap p-values come from 1,999 Rademacher-weighted bootstrap replications with the null imposed on the tested contrast. Bootstrap p-values are reported to three decimals and are bounded below by 1/2,000.

I.4 Model-Specific Frame-Manipulation Effects

Table S10 reports model-specific frame regressions for the four-model H3 subset. Each model-specific regression estimates Δ as a function of frame indicators and profile fixed effects, using the neutral frame as the reference condition. The pooled column corresponds to the main H3 specification reported in the main text, which pools the same four models and includes model and profile fixed effects.

Table S10: Model-specific frame-manipulation effects

Frame contrast	GPT-4.1	Llama 3.1 70B	Qwen3.7 Plus	Mistral Large	Pooled
Autonomy vs. neutral	8.56***	10.07***	-2.87**	4.07***	4.96***
Collective obligation vs. neutral	-6.97***	-1.78	-14.65***	-20.11***	-10.88***
Equity/access vs. neutral	8.41***	9.36***	-0.94	2.59*	4.84***

Notes: Each model-specific column reports a regression of Δ on frame indicators and profile fixed effects for the indicated model. The pooled column reports the main H3 specification for the four-model H3 subset with model and profile fixed effects. Standard errors are clustered at the model–profile level in all columns. The neutral frame is the omitted reference category. Positive coefficients indicate larger simulated burden effects relative to neutral; negative coefficients indicate smaller simulated burden effects. Standard errors and 95% confidence intervals are reported in Table S11. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table S11 reports the corresponding standard errors and confidence intervals. The model-specific results show both robustness and heterogeneity within the H3 subset. The collective-obligation frame reduces simulated burden effects in all four models (significantly in GPT-4.1, Qwen3.7 Plus, and Mistral Large; not statistically distinguishable from zero in Llama 3.1 70B Instruct). The autonomy frame increases simulated burden effects in GPT-4.1, Llama 3.1 70B Instruct, and Mistral Large, but *reverses sign* for Qwen3.7 Plus (-2.87, 95% CI $[-4.53, -1.22]$), where activating autonomy concerns reduces the simulated burden effect rather than amplifying it. The equity/access frame is positive and significant for GPT-4.1, Llama 3.1 70B Instruct, and Mistral Large, and statistically indistinguishable from zero for Qwen3.7 Plus. Mistral Large is especially informative because it has the lowest MFQ composite score in the expanded fifteen-model sample, yet all three frame manipulations shift its simulated burden effects in the expected directions. These results support the pooled H3 finding for the collective-obligation contrast while showing that model endpoints differ in both the magnitude and the sign of responsiveness to specific governance-value frames; the pooled autonomy and equity estimates are averages over a genuinely mixed set of model-level responses, and the Qwen3.7 Plus reversal is the clearest caution against reading a pooled frame effect as a property of LLMs in general.

Pair Fixed-Effects Robustness

The pooled H3 estimates were re-estimated with one fixed effect per model–profile pair (108 pairs, up to twenty assigned-frame observations each; 2,159 observations total) in place of the separate model and profile fixed effects, holding the clustering of standard errors at the model–profile level so that only the fixed-effect structure changes:

Table S11: Model-specific frame-manipulation effects with standard errors and confidence intervals

Model	Frame contrast	Coefficient	SE	95% CI
GPT-4.1	Autonomy vs. neutral	8.56***	0.76	[6.99, 10.12]
GPT-4.1	Collective obligation vs. neutral	-6.97***	0.81	[-8.63, -5.31]
GPT-4.1	Equity/access vs. neutral	8.41***	0.77	[6.83, 9.99]
Llama 3.1 70B Instruct	Autonomy vs. neutral	10.07***	1.46	[7.07, 13.08]
Llama 3.1 70B Instruct	Collective obligation vs. neutral	-1.78	1.21	[-4.27, 0.72]
Llama 3.1 70B Instruct	Equity/access vs. neutral	9.36***	1.75	[5.76, 12.97]
Qwen3.7 Plus	Autonomy vs. neutral	-2.87**	0.81	[-4.53, -1.22]
Qwen3.7 Plus	Collective obligation vs. neutral	-14.65***	1.06	[-16.84, -12.47]
Qwen3.7 Plus	Equity/access vs. neutral	-0.94	0.84	[-2.66, 0.78]
Mistral Large	Autonomy vs. neutral	4.07***	0.98	[2.05, 6.09]
Mistral Large	Collective obligation vs. neutral	-20.11***	0.89	[-21.93, -18.29]
Mistral Large	Equity/access vs. neutral	2.59*	1.13	[0.28, 4.91]
Pooled	Autonomy vs. neutral	4.96***	0.70	[3.57, 6.35]
Pooled	Collective obligation vs. neutral	-10.88***	0.84	[-12.54, -9.21]
Pooled	Equity/access vs. neutral	4.84***	0.70	[3.45, 6.24]

Notes: Each row reports a frame contrast relative to the neutral frame. Model-specific regressions include profile fixed effects. The pooled regression includes model and profile fixed effects. Standard errors are clustered at the model–profile level for the pooled column (108 clusters) and at the profile level within each model for the model-specific columns (27 clusters each), because profiles are the controlled test cases and queries within a profile are repeated measures. The pooled specification uses $N = 2{,}159$ profile–repetition observations (540 for GPT-4.1, 539 for Llama 3.1 70B Instruct, 540 for Qwen3.7 Plus, and 540 for Mistral Large, after excluding responses that failed the structured-output checks). The dependent variable is $\Delta = W^{low} - W^{high}$. Positive coefficients indicate larger simulated burden effects relative to neutral. Significance stars are based on Holm–Bonferroni-adjusted p-values: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table S12: Pooled H3 estimates under two-way and pair fixed effects

Contrast (vs. neutral)	Model + profile FE	Pair FE
Autonomy	+4.96 [+3.57, +6.35]	+4.96 [+3.56, +6.36]
Collective obligation	−10.88 [−12.54, −9.21]	−10.88 [−12.55, −9.20]
Equity/access	+4.84 [+3.45, +6.24]	+4.86 [+3.45, +6.26]

Notes: Standard errors clustered at the model–profile level (108 clusters) in both columns; all six estimates have $p < 0.001$. Because frames are assigned rather than observed, 75% of the variance in the frame indicator lies within a pair, so the pair-fixed-effects specification retains essentially the same identifying variation and the estimates are unchanged; adjusted R^2 rises from 0.775 to 0.809. The pair-fixed-effects specification provides stricter cell-specific adjustment and yields essentially identical estimates.

J Additional Robustness Checks

This appendix reports additional robustness checks for the H1 orientation–effect analysis. The main H1 specification uses the MFQ composite (individualizing minus binding foundations, standardised across endpoints) as the model-level predictor of average burden effects. Because the composite combines five theoretically distinct moral foundations with fixed directional weights, we also examine whether the foundation-specific patterns persist and whether the composite null is stable across model composition.

J.1 Caveats Qualified in the Main Text

Three caveats qualify the null H1 result. First, the MFQ composite imposes structural assumptions—equal weighting across five foundations, fixed directional signs (individualizing positive, binding negative), and an additive structure—that may not match how value dimensions interact with simulated policy effects. Second, at $N = 15$ model endpoints, model-level regressions have limited power, which is why the power and minimum-detectable-effect diagnostics below are reported alongside the coefficient estimates. Third, the fifteen endpoints are a convenience sample of commercially available models, so the estimates are descriptive of this sample rather than high-powered population estimates.

J.2 Expanded-Sample H1 Robustness

Table S13 summarizes the key H1 robustness results for the original nine-model sample and the expanded fifteen-model sample. The MFQ composite does not predict model-level burden effects in either sample. In the original nine-model sample, the composite coefficient is positive and statistically indistinguishable from zero ($\hat{\beta} = 4.55$, $p = 0.172$, $R^2 = 0.248$). After expanding to fifteen endpoints, the coefficient remains statistically indistinguishable from a null effect ($\hat{\beta} = 2.96$, 95% CI $[-2.17, 8.09]$, $p = 0.235$, $R^2 = 0.107$). Re-standardising the dimensions within the nine-endpoint subset gives a smaller positive coefficient ($\hat{\beta} = 1.92$, $p = 0.151$, $R^2 = 0.270$); the slope is scale-dependent, so both standardisations are reported.

Table S14 reports the complete set of foundation-specific regressions in the fifteen-endpoint sample: each of the five foundations is fitted separately. None of the five is individually significant (all $p \geq 0.095$). The two binding foundations show the largest point estimates in the theoretically expected (negative) direction—loyalty $\hat{\beta} = -4.73$, $p = 0.095$, $R^2 = 0.200$; authority $\hat{\beta} = -4.25$, $p = 0.138$, $R^2 = 0.161$—while the individualizing foundations are essentially flat (care $\hat{\beta} = -0.49$, $p = 0.870$; fairness $\hat{\beta} = 0.34$, $p = 0.909$). This directional pattern is consistent with the individualizing-minus-binding construction but does not reach conventional significance for any single foundation, so the dimension-level check does not overturn the composite null.

Two composite variants confirm that the null is not an artifact of the individualizing-minus-binding sign convention. An individualizing-only composite ($Z_{\text{care}} + Z_{\text{fairness}}$) has no association with burden effects ($\hat{\beta} = -0.04$, $p = 0.979$, $R^2 = 0.000$), and a leave-loyalty-out composite—dropping the strongest single foundation—remains non-significant ($\hat{\beta} = 1.22$, $p = 0.303$, $R^2 = 0.081$).

Table S13: H1 robustness across original and expanded model samples

Predictor and sample	Coefficient	p	R^2	Interpretation
MFQ composite, $N = 9$, fifteen-endpoint standardisation	4.55	0.172	0.248	Null
MFQ composite, $N = 9$, re-standardised within nine	1.92	0.151	0.270	Null
MFQ composite, $N = 15$	2.96	0.235	0.107	Null
Individualizing-only composite, $N = 15$	−0.04	0.979	0.000	Null
Leave-loyalty-out composite, $N = 15$	1.22	0.303	0.081	Null

Notes: Each row reports a separate bivariate model-level regression of $\bar{\Delta}_m$ on the listed predictor or MFQ variant. The $N = 9$ rows use the original nine-endpoint sample; the $N = 15$ rows use the expanded sample. MFQ dimensions are standardised across the sample used for aggregation (Appendix C): the first $N = 9$ row keeps the fifteen-endpoint standardisation, the second re-standardises the five dimensions within the nine-endpoint subset. Both leave the H1 conclusion unchanged, but the slope is scale-dependent (4.55 [−2.53, 11.64] versus 1.92 [−0.83, 4.68]), so the two are reported together. Higher $\bar{\Delta}_m$ indicates a larger predicted reduction in willingness under high administrative burden. The individualizing-only and leave-loyalty-out rows are reported as sensitivity diagnostics, not as validated alternative indices.

Table S14: Foundation-specific H1 regressions in the fifteen-endpoint sample

Predictor (standardised)	Coefficient	SE	95% CI	p	R^2
Care/Harm	-0.49	2.93	[-6.82, +5.84]	0.870	0.002
Fairness/Reciprocity	+0.34	2.93	[-5.99, +6.68]	0.909	0.001
Ingroup/Loyalty	-4.73	2.62	[-10.40, +0.94]	0.095	0.200
Authority/Respect	-4.25	2.69	[-10.05, +1.55]	0.138	0.161
Purity/Sanctity	-3.35	2.78	[-9.36, +2.66]	0.250	0.100
MFQ composite (reference)	2.96	2.37	[-2.17, +8.09]	0.235	0.107

Notes: Each row is a separate bivariate OLS regression of the model-level mean burden effect (points on a 0–100 willingness scale) on a single standardised MFQ-30 foundation across the fifteen endpoints. Foundations are standardised within the fifteen-endpoint sample (Appendix C); higher values indicate a stronger measured foundation. None of the five foundations is individually significant, so the dimension-level diagnostic check does not overturn the composite null; the loyalty row is one of five rather than the only dimension reported. The composite row is repeated for reference.

J.3 Leave-One-Model-Out Diagnostics

Leave-one-model-out diagnostics further support the main H1 conclusion. The composite MFQ coefficient is not a stable predictor of model-level burden effects. In the expanded sample, the bivariate MFQ association remains positive but weak and statistically indistinguishable from zero in every leave-one-model-out fit (slope range 0.72–3.79; all $p > 0.10$). Dropping Gemini 2.5 Pro shrinks the slope the most (0.72, $p = 0.824$); dropping GPT-4.1 inflates it the most (3.79, $p = 0.125$); neither fit reaches conventional significance.

These diagnostics reinforce the interpretation that H1 does not provide evidence for a stable aggregate orientation–effect relationship. The MFQ-30 is a human-validated questionnaire used here as an MFQ-30–derived probe, but the expanded analysis does not support treating a scalar moral-foundations composite as a reliable predictor of simulated burden-effect magnitude across models.

J.4 Leave-One-Profile-Out Diagnostics

We re-estimated each of the three main models twenty-seven times, excluding one synthetic profile at a time (27 fits per model). Coefficient ranges across the 27 fits are archived with the replication package. The H1 MFQ composite slope stays between 2.5 and 3.4 and is never statistically distinguishable from zero (p between 0.16 and 0.33), so the H1 null is not produced by any single test case. For H2 under the primary pair-fixed-effects specification, no retained frame is statistically distinguishable from zero in any of the 27 fits: the access-barrier coefficient is positive in 25 of 27 fits (range -0.09 to $+0.32$), and the coercive-backlash coefficient is positive in all 27 fits ($+0.37$ to $+2.33$) but never significant. (Collective responsibility is excluded from the primary specification because corrected coding leaves only one positive rationale; see Appendix I.) For H3, all three frame contrasts retain both sign and significance in all 27 fits (autonomy $+4.64$ to $+5.18$; collective obligation -11.01 to -10.70; equity/access $+4.56$ to $+5.10$; $p < 0.001$ throughout).

J.5 Power and Minimum Detectable Effect Diagnostics

The original nine-model H1 analysis was underpowered for small or moderate model-level associations. In that sample, the approximate 80% minimum detectable slope for the composite MFQ index was $\beta \approx 10.2$ (5.7 in the fifteen-endpoint sample). These thresholds are expressed on the standardised-composite scale used in the main analysis; they correspond to roughly 3.2 and 3.0 standard errors for the nine- and fifteen-endpoint samples, and the archived power simulation is included in the replication materials. The observed MFQ coefficient (2.96) is well within the confidence interval and does not reach the MDE, indicating the sample was underpowered to detect an effect of this magnitude. Expanding the model sample to fifteen endpoints improves coverage but does not eliminate the basic limitation that H1 relies on model-level variation.

For this reason, the H1 estimates should be interpreted as descriptive diagnostics rather than high-powered population estimates. The expanded results are most informative as evidence against a large linear association between the composite MFQ index and simulated burden effects in this model sample. They should not be read as evidence that moral foundations are irrelevant to LLM behavior. Instead, the strongest evidence in the paper comes from the randomized within-model frame-manipulation experiment;

H2 shows that naturally expressed rationale frames do not provide a robust within-pair diagnostic.